

Modeling wind turbine blades by geometrically-exact beam and shell elements: A comparative approach

Celso Jaco Faccio Júnior*, Ana Carolina Pegoraro Cardozo, Valdemar Monteiro Júnior, Alfredo Gay Neto*

Department of Structural and Geotechnical Engineering – Polytechnic School at University of São Paulo, Brazil

ARTICLE INFO

Keywords:

Geometrically-exact
Beam
Shell
Wind turbine
Blade

ABSTRACT

Novel designs of wind turbine blades may lead to very flexible structures experiencing large deformation not only in extreme events but also on operational conditions. In this context, this work aims to compare two geometrically nonlinear structural modeling approaches that handle large deformation of blade structures: 3D geometrically-exact beam and shell finite element models. Specifically, the focus is on studying the accuracy limits of the beam model when compared to the shell model in scenarios of large deformations. Regarding the beam model, due to geometric complexity of typical cross-sections of wind turbine blades, a theory is adopted that allows for the creation of arbitrary multicellular cross-sections. Two simplified blade geometries are considered, and comparisons between the models are made in statics and dynamics. Results showed that the beam and shell models present very similar global behavior. Locally, however, extreme load cases led shell models to show local buckling phenomena close to the trailing edge, which may play a role in novel HAWT designs.

1. Introduction

The total wind power capacity installed in the world increased approximately 17% in 2015, rising from 369 GW to 432 GW, as presented by the Global Wind Energy Council (GWEC) [1]. Even with possibilities of novel conceptions for wind turbines, a very important and still most often used model is the horizontal axis wind turbine (HAWT) [2]. Throughout its history, HAWT has significantly evolved from a power of 25 kW, with a rotor diameter of 10 m, to 6 MW with a rotor diameter of 164 m [3]. Recent studies for a 10 MW offshore HAWT [4] are being conducted.

In this context, the main reason for the increasing size of wind turbines is that their power capacity is directly related to the area covered by the blades, and therefore, related to the square of their length. However, the increase of the length of blades is not a trivial engineering process. Assumptions and models previously used for short blades may be inappropriate for the new proposals. In fact, slender and flexible wind turbine blades may experience high nonlinear structural behavior. Furthermore, aeroelasticity may play a role. As a consequence, considerable research is being employed to produce realistic wind turbine blade models.

There are basically three main approaches to establishing structural numerical models for analysis of wind turbine blades. When there is an interest in local structural issues, a 3D model composed of solid finite elements provides extreme details for stress distributions, in detriment of a very high computational cost. However, it usually cannot be used to model structural global behavior. For that, a 3D model based on shell finite elements may be proposed. It simplifies local details, but usually is also a good compromise to analyze overall structural behavior, still predicting important phenomena such as local buckling. A third approach, the computationally cheapest one, is based on 3D beam structural assumptions. It may be used as a very efficient approach for the initial design of wind turbine blades. The focus of the present work is on the comparison of shell and beam models, which are discussed next.

Discretization of wind turbine blades using 3D composite shell elements may be very accurate. This kind of model provides the possibility to conduct different studies, including thickness variation and/or anisotropic nonhomogeneous materials. The simulation of detailed shell structures, however, may take a considerable amount of time, sometimes prohibitive for initial design studies of wind turbine blades. However, the growth of computational power is providing more possibilities for the use of such models. Some contributions regarding 3D

* Corresponding authors.

E-mail addresses: celsojf@usp.br (C.J. Faccio Júnior), ana.cardozo@usp.br (A.C.P. Cardozo), valdemar.moj@gmail.com (V. Monteiro Júnior), alfredo.gay@usp.br (A. Gay Neto).

<https://doi.org/10.1016/j.engstruct.2018.09.032>

Received 20 March 2018; Received in revised form 31 July 2018; Accepted 13 September 2018

Available online 26 November 2018

0141-0296/ © 2018 Elsevier Ltd. All rights reserved.

shell FEM analysis for modeling of wind turbines are briefly commented on below. An alternative free open source tool to make 3D finite element models (FEM) meshes of wind turbine blades is the Numerical and Manufacturing Design Tool (NuMAD) developed by [5]. In [6], a coupling of a blade shell FEM model built on Abaqus™ [7] is presented, with the blade element momentum theory of the program WT_Perf [8]. In [9], the aerodynamic loads are investigated by coupling a Navier-Stokes computational fluid dynamics (CFD) solver with a FEM-based computational structural dynamics solver. In [10], a full wind turbine, including the nacelle and tower, is simulated with fully-coupled 3D fluid-structure interaction. In [11], the bend-twist coupling effect is investigated regarding fatigue loads and flutter instability. In [12], the fluid-structure interaction at a full-scale wind turbine blade is investigated in ANSYS™ [13]. In [14], a physics-based multi-scaled progressive damage model for predicting the durability of wind turbine blade structures is presented. In [15], a structural optimization model of wind turbine blades using finite element analysis and a genetic algorithm is developed. In [16] a series of solutions are presented, based in shell model analysis. Trailing edge local buckling is addressed.

Even with many contributions previously mentioned addressing high-hierarchy models, the discretization of wind turbines into 3D beam-like structural models has proven to be computationally efficient. Indeed, the blade and the tower of wind turbines are usually slender structures. Once a beam-like model is adopted, three possible modeling strategies may be taken, i.e., a modal approach, a multi-body dynamics (MBD) method, and a FEM method [17]. The modal approach consists of modeling the wind turbine dynamics by a combination of previously known modal shapes. This method is computationally efficient since it greatly reduces the number of degrees of freedom (DOF's). However, there is a drawback that it is limited to the available eigen-modes. Furthermore, if the structure presents nonlinear behavior, the advantageous simplicity of the method is no more kept. On MBD modeling, the structure is discretized into a simplified system composed of rigid and flexible parts, interconnected by forces, springs, dashpots, and kinematic constraints. Finally, the classical FEM approach discretizes the wind turbine structure into segments or elements interconnected by nodes. Although the slightly higher computational cost, the classical 3D FEM wind turbine modeling, composed of beam elements, allows the use of linear or nonlinear formulations.

Concerning the beam theories adopted in wind turbine models, geometrically linear and nonlinear theories have been used to describe the structural behavior. Regarding wind turbine blades, classical beam theories were initially widely used to model them, particularly due to the higher stiffness of these structures when compared to helicopter blades [18]. However, as mentioned in [18], nonlinear beam theories may become necessary according to the wind turbines flexibility. Most recently, the use of geometric nonlinear beam theories in the context of slender and flexible wind turbine blades were revealed not as a possibility, but as a reality and a trend [17,19]. In this context, several studies have been developed to optimize the design of wind turbine blades [20,21]. Beam models are very welcome since they are computationally-light, which is essential for optimizing practical procedures. The HAWC2 (Horizontal Axis Wind Turbine Simulation Code 2nd generation) developed by the DTU (Technical University of Denmark) is a nonlinear aeroelastic model based on a MBD formulation and Timoshenko beam elements [22]. In this code, the nonlinear effects depend on a proper subdivision of the blade into a number of bodies [22,23]. In [24], the authors develop a model called NAM_WTB (Nonlinear Aeroelastic Model for Wind Turbine Blades) that couples a geometrically exact beam theory formulation developed by [25] and the well-known blade element momentum (BEM) model for aerodynamic loads. In [26], the authors present a code called BeamDyn that implements the geometrically exact beam theory inside the modular framework FAST.

Among the mentioned high-hierarchy shell models and the quick-solving beam-like models, a natural question that needs to be answered is regarding the comparative modeling aspects of both approaches. Strong and weak characteristics of each method may be comparatively discussed, leading to clarity on the engineering limitations of each model. In this context, the objective of this article is to compare the predicted physical behavior of a wind turbine blade, when modeled using: (a) a 3D geometrically-exact beam FEM model, and (b) a 3D geometrically-exact shell FEM model. The intention is to verify the accuracy range of the nonlinear beam formulation proposed, taking the shell model as a more complete and realistic reference. For that, airfoil shapes (including webs) with realistic dimensions are considered for the blades. The main contribution of this article is in the comparison of the wind turbine blade beam and shell models using the geometrically-exact formulation. In the geometrically-exact formulation, each beam or shell element is not limited to small displacement or small rotation, being therefore intrinsically geometrically nonlinear.

As this article focus on the nonlinear responses of the structural models, the elastic properties of some laminated composite materials are calculated by a simplified method. It is important to highlight some essential features of the geometrically-exact beam model adopted in this article such as the general constitutive matrix regardless of the chosen beam axis; the bend-twist coupling consistently obtained considering the webs contribution and the equivalent material properties calculated with the aid of the Composite Laminate Theory. The combination of these features as proposed in the present article, as far as the authors' knowledge goes, has not yet been used in the modeling of wind turbine blades.

In order to improve the richness, discussions, and applicability of the comparisons proposed here, two distinct blade geometries and two kinds of structural analysis are performed. Regarding the blade geometry, an initial analysis proposes a constant cross-section along the blade length, while a more elaborated model proposes a linear variation of the profile chord along the blade length. The structural analysis consists of static and transient dynamics simulations and, in both cases, the structure experiences geometric nonlinear behavior. The article closes with a comparative discussion of all approaches concerning statics, dynamics, and influence of geometric nonlinear effects on blade structural behavior.

A brief description of each section of this article is presented as follows. First, the geometrically-exact beam model formulation is introduced, with emphasis given to the constitutive matrix derivation. Next, the geometrically-exact shell model adopted is briefly presented. After, the computational background used on the simulations is presented, including (1) an original computer-aided design (CAD) tool *WindTurbine* [27] to evaluate the beam constitutive matrix and (2) the finite element solver (*Giraffe* [28,29]). Then, the proposed case studies are presented by comparing the two modeling approaches. Discussions on comparative results of both models are presented. Finally, recommendations are made as part of the concluding remarks.

2. Geometrically-exact beam model

Geometrically-exact beam structural models may be simply described as a strategy to decompose the beam deformation in two main effects: (a) a general rigid-body movement of the whole cross-section, and (b) a local deformation of each cross-section (such as warping or shape changes). The effect (a) considers that the movement of each cross-section is not limited to small deflections or small rotations. This will lead to a convenient technique to model parts moving and experiencing large displacements and finite rotations. The geometrically-exact beam theory is able to naturally describe the kinematics of mechanisms, such as helicopter blades [30], offshore risers [31–33], thin-walled beams [34–36], and wind turbines [24]. Recently, the

geometrically-exact beams have been used together with joint constraints, providing a real possibility to simulate mechanisms, as an alternative to other MBD techniques (see, e.g., [37–40]).

The initial developments regarding the geometrically-exact formulation adopted in this article were presented in [41], where the author shows a 3D beam formulation, based on the original plane formulation proposed by [42]. Both works may be seen as more general than the Kirchhoff-Love's rod model [43], which does not consider shear deformation. It is remarkable that many authors dedicated efforts in order to improve the geometrically-exact beam theory by incorporating warping [35] and providing a general-axis description, which is one of the bases of the present work [44,45].

In the present work, the aim is not at covering the geometrically-exact theory details broadly, but to focus on the particular challenges when trying to solve wind turbine blade structural analysis using such theory. Thus, some studies related to background on geometrically-exact beam theories formulations adopted in this article are briefly commented as follows, where the reader may find theoretical details. In [46] the author presents a full nonlinear 3D beam model that incorporates transverse shear and torsion-warping deformation. Furthermore, a class of reduced elastic constitutive equations is also presented. In [44] the rotation of 3D geometrically-exact beams is formulated using the Euler-Rodrigues formula, while in [47] the rotation is expressed using only Rodrigues parameters. In [48] the dynamic formulation of a geometrically-exact beam is used to simulate an offshore riser structures with hydrodynamic loads and contact interaction with the seabed, which is the basis here employed in our analysis, with modifications stated next in constitutive matrix.

2.1. Constitutive matrix for general beam cross-sections with torsional effects

The constitutive matrix formulation presented below is based on [46], where a class of reduced elastic constitutive equations is presented. Moreover, differently from the beam formulation from [46] – that has eight internal loads contributions, the adopted formulation presents six internal loads, such as done in [45]. A constitutive matrix, for a particular structural model, is a mathematical entity that relates the generalized stresses and generalized strains. It is desirable for the constitutive matrix to be able to capture possible couplings between internal loads, due to the arbitrary material distribution at the cross-section. One important aspect, usually representing a concern for beam models, is the choice of the beam axis position. According to such choice, distinct coupling terms will take place in the constitutive matrix, leading even to possible changes in the internal loads interpretation. Our choice in the present work is for a completely general axis position, not necessarily lying at the barycenter, shear center, or other particular point of the cross-section. This makes the derivation more elaborated, but afterwards creates a desirable generality to construct beam models with variable cross-sections along length. This scenario is very expected in the context of wind turbine blades that usually present complex geometries along length.

Some geometric properties and identities are now defined, according to the notation presented in Fig. 1. The cross-section area is

$$A = \int_A dA = \int_A dx_1 dx_2 \quad (1)$$

The first order moments of inertia are

$$S_1 = \int_A x_2 dA \text{ and} \quad (2)$$

$$S_2 = - \int_A x_1 dA \quad (3)$$

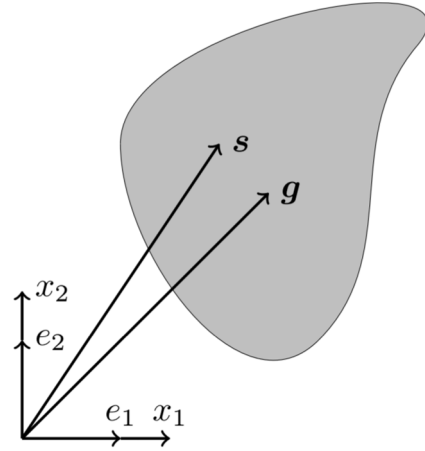


Fig. 1. Cross-section notation adopted.

The second order moments of inertia, product of inertia, and polar moment of inertia are respectively given by

$$I_1 = \int_A x_2^2 dA \quad (4)$$

$$I_2 = \int_A x_1^2 dA \quad (5)$$

$$I_{12} = - \int_A x_1 x_2 dA \text{ and} \quad (6)$$

$$I_0 = \int_A (x_1^2 + x_2^2) dA = I_1 + I_2 \quad (7)$$

It is also convenient to define the cross-section vector positions for some points of interest, such as the centroid g and the shear center s . They are both defined in each cross-section reference plane by:

$$g = g_1 e_1 + g_2 e_2 \text{ and} \quad (8)$$

$$s = s_1 e_1 + s_2 e_2 \quad (9)$$

with

$$g_1 = -\frac{S_2}{A} \text{ and} \quad (10)$$

$$g_2 = \frac{S_1}{A} \quad (11)$$

In contrast, the process of obtaining the shear center coordinates that will rule the bend-twist coupling of the beam model may not be immediately obtained. It is possible, however, to express the shear center coordinates with respect to a chosen point of interest (e.g., the beam axis position in cross-section plane) as

$$s_1 = \frac{1}{n_2} \int_A (r_1 q_2 - r_2 q_1) dA \text{ and} \quad (12)$$

$$s_2 = -\frac{1}{n_1} \int_A (r_1 q_2 - r_2 q_1) dA \quad (13)$$

where n_1 and n_2 are parallel forces applied respectively in directions of local axes e_1 and e_2 , r_1 and r_2 are the components of a vector that reads each cross-section point of interest and q_1 and q_2 are the components of the shear stress function. A more detailed discussion about the shear center may be found in [49–51].

It is now convenient to define first order moments of inertia with respect to the shear center with the aid of the St. Venant warping function ψ^s , also relative to the shear center, as

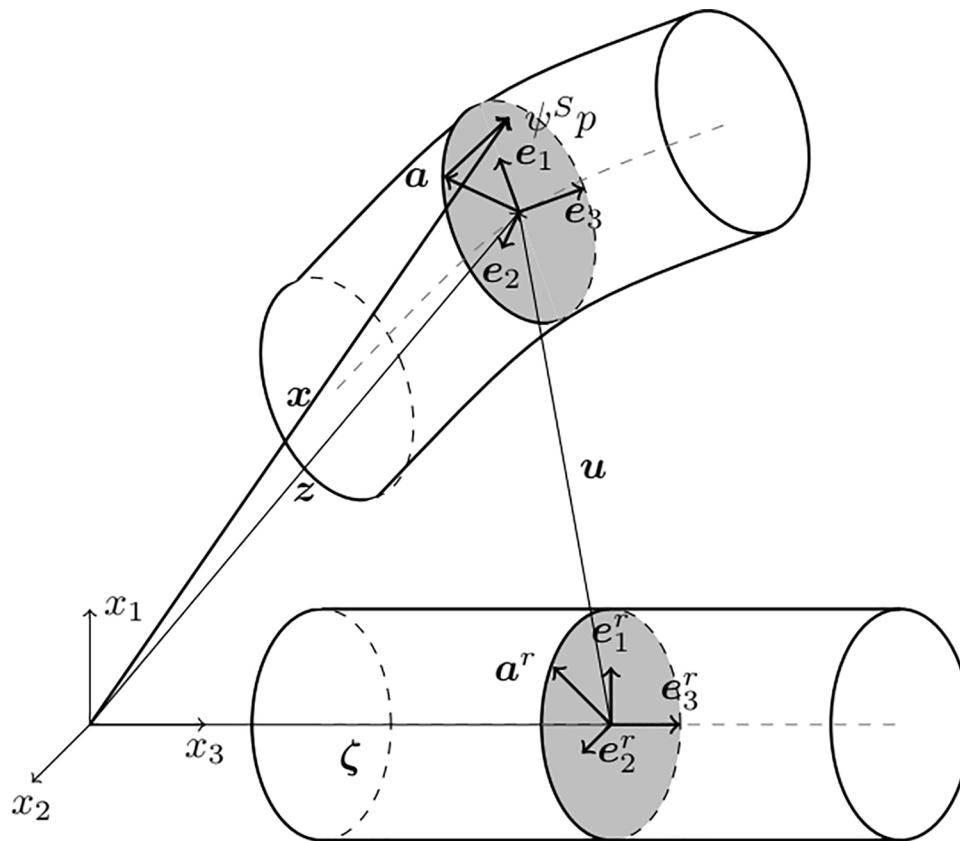


Fig. 2. Kinematic hypothesis of the proposed beam model.

$$\int_A \psi_1^S dA = \int_A (x_2 - s_2) dA = S_1^S = S_1 - As_2 \text{ and} \quad (14)$$

$$\int_A \psi_2^S dA = - \int_A (x_1 - s_1) dA = S_2^S = S_2 + As_1 \quad (15)$$

where the superscript $(\cdot)^S$ denotes from now on calculations with respect to the shear center coordinates and the notation $(\cdot)_{,i}$ represents the derivative of a quantity with respect to coordinate x_i . A detailed discussion about the St. Venant warping functions may be found in [34,47,52,53].

Other useful identities related to the shear center and the St. Venant warping function ψ^S as presented in [34,45,47] are

$$\int_A (x_1 \psi_2^S - x_2 \psi_1^S) dA = I_t^S - I_0 + A(g_1 s_1 + g_2 s_2) \quad (16)$$

$$\int_A (\psi_1^S \psi_1^S + \psi_2^S \psi_2^S) dA = I_0^S - I_t^S \quad (17)$$

$$I_0^S = I_0 - 2A(g_1 s_1 + g_2 s_2) + A(s_1^2 + s_2^2) \quad (18)$$

where I_t^S is the St. Venant torsional inertia, given by the relation between a pure torque T applied to the cross-section, a constant or homogenized Shear Modulus G and a rate of twist ϕ , such that

$$I_t^S = \frac{T}{G\phi} \quad (19)$$

In order to establish a constitutive matrix, one has to assume a displacement field (kinematic hypothesis) that describes the movement of each material point of the model. From this, one has to calculate the deformation gradient and the corresponding strain tensor according to the assumption regarding possible infinitesimal strains (as assumed

here). Next, a material law, for example, the Hooke's law (elastic linear hypothesis) may be used to associate the stresses and strains.

Consider now that the position vector of a material point $\mathbf{x}$ of a given beam, as presented in Fig. 2, may be expressed by

$$\mathbf{x} = \zeta \mathbf{e}_3^r + \mathbf{u} + Q\mathbf{a}^r + \psi^S p \mathbf{e}_r^3 \quad (20)$$

where $\mathbf{u}$ is the displacement of the beam axis, $\mathbf{Q}$ is the rotation tensor, describing the current orientation of cross-sections, $\mathbf{a}'$ is a vector that reads all points at the beam section ($\mathbf{a}' = x_1 \mathbf{e}_1' + x_2 \mathbf{e}_2'$), and the term $\psi^s p e_i^s$ describes out of plane warping displacements of the beam cross-section. It is essential for the formulation to note that the displacement vector $\mathbf{u}$ and the rotation tensor $\mathbf{Q}$ are, respectively, not limited to small displacements and small rotations.

Note that, at this kinematic hypothesis, the out of plane displacement is described not only by the warping function ψ^S but also by a new parameter p . This notation defines the out of plane displacement of a beam section as a multiplication of a warping function ψ^S ; in this context, $\psi^S = \hat{\psi}(x_1, x_2)$ is the shape of the warping displacement field and a scalar intensity factor p .

The rotation tensor may be written using Rodrigues' rotation vector $\alpha = [\alpha_1 \ \alpha_2 \ \alpha_3]^T$, or $\alpha = \alpha \mathbf{e}$, with $\mathbf{e}$ representing the rotation axis direction experienced by each cross-section, being $\alpha = 2 \tan(\theta/2)$. The scalar θ is the magnitude of the Euler rotation vector associated with the cross-section. It is still possible to define the skew-symmetric tensor $\mathbf{A}$ by

$$A = \begin{bmatrix} 0 & -\alpha_3 & \alpha_2 \\ \alpha_3 & 0 & -\alpha_1 \\ -\alpha_2 & \alpha_1 & 0 \end{bmatrix} \quad (21)$$

Then, it is possible to write a linear approximation to the rotation tensor $\mathbf{Q}$, valid for small rotations, given by $\mathbf{Q} \approx \mathbf{I} + \mathbf{A}$. With these definitions, Eq. (20) may be rewritten as

$$\mathbf{x} = \zeta \mathbf{e}_3^r + \mathbf{u} + (\mathbf{I} + \mathbf{A})(x_1 \mathbf{e}_1^r + x_2 \mathbf{e}_2^r) + \psi^S p \mathbf{e}_3^r \quad (22)$$

and by consequence, the deformation gradient $\mathbf{F}$ of $\mathbf{x}$ is given as

$$\mathbf{F} = \mathbf{x}_{,1} \otimes \mathbf{e}_1^r + \mathbf{x}_{,2} \otimes \mathbf{e}_2^r + \mathbf{x}' \otimes \mathbf{e}_3^r \quad (23)$$

where $(\cdot)'$ is the partial derivatives with respect to ζ . Moreover, using Vlasov's hypothesis [53] that the warping displacement is proportional to the rotation per unit length of the beam axis ($p = \alpha' \cdot \mathbf{e}_3^r = \alpha'_3$), it is possible to rewrite $\mathbf{F}$ as

$$\mathbf{F} = \mathbf{I} + (\mathbf{A}\mathbf{e}_1^r + \psi_{,1}^S \alpha'_3 \mathbf{e}_3^r) \otimes \mathbf{e}_1^r + (\mathbf{A}\mathbf{e}_2^r + \psi_{,2}^S \alpha'_3 \mathbf{e}_3^r) \otimes \mathbf{e}_2^r + (\mathbf{u}' + \mathbf{A}'\mathbf{a}' + \psi^S \alpha'_3 \mathbf{e}_3^r) \otimes \mathbf{e}_3^r \quad (24)$$

From the deformation gradient $\mathbf{F}$, it is possible now to calculate a strain measure $\nabla \mathbf{S}$ by

$$\nabla \mathbf{S} = \mathbf{F} - \mathbf{I}, \text{ such that} \quad (25)$$

$$\nabla \mathbf{S} = (\mathbf{A}\mathbf{e}_1^r + \psi_{,1}^S \alpha'_3 \mathbf{e}_3^r) \otimes \mathbf{e}_1^r + (\mathbf{A}\mathbf{e}_2^r + \psi_{,2}^S \alpha'_3 \mathbf{e}_3^r) \otimes \mathbf{e}_2^r + (\mathbf{u}' + \mathbf{A}'\mathbf{a}' + \psi^S \alpha'_3 \mathbf{e}_3^r) \otimes \mathbf{e}_3^r \quad (26)$$

The strain measure $\nabla \mathbf{S}$ yields to

$$\nabla \mathbf{S} = \begin{bmatrix} 0 & -\alpha'_3 & u'_1 - \alpha'_3 x_2 \\ \alpha'_3 & 0 & u'_2 + \alpha'_3 x_1 \\ -\alpha_2 + \psi_{,1}^S \alpha'_3 & \alpha_1 + \psi_{,2}^S \alpha'_3 & u'_3 + \alpha'_1 x_2 - \alpha'_2 x_1 + \psi^S \alpha'_3 \end{bmatrix} \quad (27)$$

Assuming that, for constitutive matrix derivation purposes, only small strains take place, one may write the infinitesimal Green-Lagrange strain tensor $\mathbf{E}$, expressed as

$$\mathbf{E} = \frac{1}{2}(\nabla \mathbf{S} + \nabla \mathbf{S}^T) \quad (28)$$

and by consequence:

$$\mathbf{E} = \begin{bmatrix} 0 & 0 & \frac{1}{2}(u'_1 - \alpha'_3 x_2 - \alpha_2 + \psi_{,1}^S \alpha'_3) \\ 0 & \frac{1}{2}(u'_2 + \alpha'_3 x_1 + \alpha_1 + \psi_{,2}^S \alpha'_3) \\ \text{Sym.} & u'_3 + \alpha'_1 x_2 - \alpha'_2 x_1 + \psi^S \alpha'_3 \end{bmatrix} \quad (29)$$

It is important to note that the current strain tensor $\mathbf{E}$ presents a degree of freedom associated with the term α'_3 . This term, which may be seen as a second order rotation with respect to the beam axis, is indeed associated with non-uniform torsion and consequently bishear and bi-moment loads. Non-uniform torsion effects are especially relevant on beams with open thin-walled cross-sections that usually present low torsion stiffness. Therefore, a reasonable hypothesis, in the context of wind turbine blades composed of multicellular closed sections, is to assume negligible non-uniform torsion effects.

From this point on, it is convenient to define two generalized strain vectors with important meanings in the context of the beam kinematics. A vector $\boldsymbol{\eta}$ defined as

$$\boldsymbol{\eta} = \mathbf{u}' - \boldsymbol{\alpha} \times \mathbf{e}_3^r = \begin{bmatrix} u'_1 - \alpha_2 \\ u'_2 + \alpha_1 \\ u'_3 \end{bmatrix} \quad (30)$$

that may be interpreted as a compact notation of the beam axial strain. And a vector $\boldsymbol{\kappa}$ defined as

$$\boldsymbol{\kappa} = \boldsymbol{\alpha}' = \begin{bmatrix} \alpha'_1 \\ \alpha'_2 \\ \alpha'_3 \end{bmatrix} \quad (31)$$

that may be interpreted as the rotation derivatives, with respect to the beam axis directions, thus containing information on curvature and twist (also called specific rotation vector).

As a result, the infinitesimal strain tensor $\mathbf{E}$ may be rewritten assuming negligible non-uniform torsion effects with the convenient strain notation of Eqs. (30) and (31) and technical notation as

$$\boldsymbol{\gamma} = \begin{bmatrix} \gamma_{13} \\ \gamma_{23} \\ \varepsilon_{33} \end{bmatrix} = \begin{bmatrix} \eta_1 - \kappa_3 x_2 + \psi_{,1}^S \kappa_3 \\ \eta_2 + \kappa_3 x_1 + \psi_{,2}^S \kappa_3 \\ \eta_3 + \kappa_1 x_2 - \kappa_2 x_1 \end{bmatrix} \quad (32)$$

In addition, considering homogeneity of material distribution and the general Hooke's law, the stress-strain relation of the beam model is

$$\boldsymbol{\sigma} = \begin{bmatrix} \sigma_{13} \\ \sigma_{23} \\ \sigma_{33} \end{bmatrix} = \begin{bmatrix} G\gamma_{13} \\ G\gamma_{23} \\ E\varepsilon_{33} \end{bmatrix} \quad (33)$$

where G and E are the material Shear Modulus and Young Modulus, respectively. Eq. (33) presents all stress components acting at any point of the beam cross-section with respect to the generalized strain vectors $\boldsymbol{\eta}$ and $\boldsymbol{\kappa}$. Therefore, the equivalent (homogenized) forces at each direction may be obtained by integration as

$$\mathbf{n} = \int_A \sigma_{13} d\mathbf{A} \mathbf{e}_1^r + \int_A \sigma_{23} d\mathbf{A} \mathbf{e}_2^r + \int_A \sigma_{33} d\mathbf{A} \mathbf{e}_3^r \quad (34)$$

or, in an explicit way, as

$$n_1 = \int_A \sigma_{13} dA = G \int_A (\eta_1 - \kappa_3 x_2 + \psi_{,1}^S \kappa_3) dA = G A \eta_1 - G S_1 \kappa_3 + G S_1^S \kappa_3 \quad (35)$$

$$n_2 = \int_A \sigma_{23} dA = G \int_A (\eta_2 + \kappa_3 x_1 + \psi_{,2}^S \kappa_3) dA = G A \eta_2 - G S_2 \kappa_3 + G S_2^S \kappa_3 \quad (36)$$

$$n_3 = \int_A \sigma_{33} dA = E \int_A (\eta_3 + \kappa_1 x_2 - \kappa_2 x_1) dA = E A \eta_3 + E S_1 \kappa_1 + E S_2 \kappa_2 \quad (37)$$

Note that identities of Eqs. (14) and (15) are applied to Eqs. (35) and (36), generating the terms S_1^S and S_2^S . These, together with the term involving S_1 and S_2 , are responsible for the coupling between the shear forces and the moment in direction $\mathbf{e}_3^r$, depending on the geometry of the cross-section and the chosen axis.

It is also possible to develop a vector of equivalent moments from integration of stresses acting at the cross-section, considering the axis position as the pole (torsion moment). Therefore, a vector of equivalent moments $\mathbf{m}$ may be obtained by integration as

$$\mathbf{m} = \int_A x_2 \sigma_{33} d\mathbf{A} \mathbf{e}_1^r - \int_A x_1 \sigma_{33} d\mathbf{A} \mathbf{e}_2^r + \int_A ((x_1 \sigma_{23} - x_2 \sigma_{13}) + (\psi_{,1}^S \sigma_{13} + \psi_{,2}^S \sigma_{23})) d\mathbf{A} \mathbf{e}_3^r \quad (38)$$

or, in an explicit way, as

$$m_1 = \int_A x_2 \sigma_{33} dA = E \int_A (x_2 \eta_3 + \kappa_1 x_2^2 - \kappa_2 x_1 x_2) dA = E S_1 \eta_3 + E I_1 \kappa_1 + E I_{12} \kappa_2 \quad (39)$$

$$m_2 = - \int_A x_1 \sigma_{33} dA = -E \int_A (x_1 \eta_3 + \kappa_1 x_1 x_2 - \kappa_2 x_1^2) dA = E S_2 \eta_3 + E I_{12} \kappa_1 + E I_2 \kappa_2, \text{ and} \quad (40)$$

$$m_3 = \int_A ((x_1 \sigma_{23} - x_2 \sigma_{13}) + (\psi_{,1}^S \sigma_{13} + \psi_{,2}^S \sigma_{23})) dA \quad (41)$$

It is interesting to observe that the component m_3 is composed of a torsion moment and a new internal moment term, coming from warping, defined as bishear moment. Thus, it is convenient to split m_3 into two contributions

$$m_3 = T + Q \quad (42)$$

where the torsion moment T is

$$T = \int_A (x_1 \sigma_{23} - x_2 \sigma_{13}) dA \quad (43)$$

and the bishear moment Q is

$$Q = \int_A (\psi_{,1}^S \sigma_{13} + \psi_{,2}^S \sigma_{23}) dA \quad (44)$$

The expansion of Eq. (43) and use of the identity of Eq. (16) leads to

$$T = \int_A (x_1 \sigma_{23} - x_2 \sigma_{13}) dA = -GS_2 \eta_2 - GS_1 \eta_1 + G(I_t^S + A(g_1 s_1 + g_2 s_2)) \kappa_3 \quad (45)$$

While the expansion of Eq. (44) and use of identities of Eqs. (16) and (17) leads to

$$Q = \int_A (\psi_{,1}^S \sigma_{13} + \psi_{,2}^S \sigma_{23}) dA = GS_1^S \eta_1 + GS_2^S \eta_2 + G(I_0^S - I_0 + A(g_1 s_1 + g_2 s_2)) \kappa_3 \quad (46)$$

Furthermore, it is possible now to calculate explicitly the equivalent moment m_3 as the sum of the T and Q contributions. Introducing then the convenient notation of $I_t = I_t^S + A(s_1^2 + s_2^2)$ and using the identity of Eq. (18), it is possible to express the m_3 component as

$$m_3 = G(S_1^S - S_1) \eta_1 + G(S_2^S - S_2) \eta_2 + GI_t \kappa_3 \quad (47)$$

Finally, it is possible to compress Eqs. (35)–(37), (39), (40) and (47) into a matrix form as

$$\begin{bmatrix} n_1 \\ n_2 \\ n_3 \\ m_1 \\ m_2 \\ m_3 \end{bmatrix} = \begin{bmatrix} GA & 0 & 0 & 0 & 0 & G(S_1^S - S_1) \\ 0 & GA & 0 & 0 & 0 & G(S_2^S - S_2) \\ 0 & 0 & EA & ES_1 & ES_2 & 0 \\ 0 & 0 & ES_1 & EI_1 & EI_{12} & 0 \\ 0 & 0 & ES_2 & EI_{12} & EI_2 & 0 \\ G(S_1^S - S_1) & G(S_2^S - S_2) & 0 & 0 & 0 & GI_t \end{bmatrix} \begin{bmatrix} \eta_1 \\ \eta_2 \\ \eta_3 \\ \kappa_1 \\ \kappa_2 \\ \kappa_3 \end{bmatrix} \quad (48)$$

It is important to note that despite all of the hypotheses (small displacements and small rotations, elastic linear material by the Hooke's law and negligible non-uniform torsion), the obtained constitutive matrix considers consistently geometric couplings for any arbitrary axis position.

3. Geometrically-exact shell model

A shell may be defined as a structure that presents one dimension (thickness) much smaller than the other two dimensions. Moreover, shell structures, from nature or manufactured, are easily found in daily life. Some few examples of shell structures are eggshells, skulls, bells, pipes, cans, wind turbine blades, among others. In fact, the wide use of shell structures is mainly due to their structural efficiency associated with bending-stretching coupling [54].

Regarding the current structural use of shell theories in engineering problems, two main theories have been widely used, the Kirchhoff-Love shell theory and the Mindlin-Reissner shell theory. The Kirchhoff-Love shell theory, often associated with the Bernoulli-Euler beam theory, was first presented by Love [55] who used the Kirchhoff assumptions [56] for their plate-bending theory. One main characteristic of the Kirchhoff-Love shell theory is that the shear deformations are neglected, resulting in a shell director always normal to the deformed shell mid-surface (the shell director is a unit vector initially orthogonal to the shell reference configuration $\mathbf{e}_3^r$ that follows the rotation of each material point of the shell mid-surface and turns into $\mathbf{e}_3$). On the other hand, the Mindlin-Reissner second order shell theory presented by Reissner [57], often

associated with the Timoshenko beam theory, incorporates shear deformations on the formulation. Therefore, unlike the Kirchhoff-Love shell theory, the Mindlin-Reissner shell theory presents a shell director that may not be orthogonal to the deformed shell mid-surface.

However, any shell theory may be roughly categorized into geometrically linear (see for e.g., [58–60]) or geometrically nonlinear (see for e.g., [61–63]). In this context, the present work adopts a Mindlin-Reissner nonlinear geometrically-exact shell theory introduced by Simo in [64]. Moreover, the models are discretized according to the fully nonlinear shell finite element T6-3i as presented in [65]. Finally, this article adopts the dynamic geometrically-exact formulation presented in [54] for the analysis of creased shell domes. Fig. 3 presents the kinematic description of the Mindlin-Reissner shell model adopted.

4. Computational implementations used for simulations

4.1. WindTurbine: a computer aided design (CAD) tool for wind turbines

To perform our study, a CAD tool was developed. The main objective of the originally developed *WindTurbine* software is to aid in the calculation of geometric properties and in the geometry definition of wind turbine blades' cross-sections. Particularly, one of the main features of the *WindTurbine* is the calculation of the geometric properties according to the formulation of Eq. (48).

In the *WindTurbine* tool, a cross-section profile is simply defined by a sequence of counterclockwise coordinates, thickness values and material elastic properties. Alternatively, one may add or remove profiles to the wind turbine blade, include webs in profiles by coordinates labels, set profiles chord and twist angle, set a reference point from where all geometric properties are calculated (beam axis), among other possibilities.

Regarding the geometric properties calculations, it is important to observe that the *WindTurbine* tool assumes that the cross-section profile is composed of flat thin-walled parts. Although there is no clear definition about the aspect ratio of a thin-walled structure, according to [50] the approximations are reasonable accurate for profiles with

$$\frac{t_{max}}{b} < 0.1 \quad (49)$$

where t_{max} is the maximum profile thickness and b is a typical profile dimension. Additionally, a main consequence of the closed multicellular thin-walled profile hypothesis is that the shear stresses distribution may be assumed constant across each profile thickness. This assumption allows the *WindTurbine* to calculate the torsional inertia and the shear center of thin-walled profiles including the webs effects in the shear stresses distribution.

In the context of a wind turbine profile that presents no webs (single cell closed profile) the torsional inertia may be obtained simply by the classical second Bredt-Bartho formula. However, when the wind turbine profile presents one or more webs (multicellular closed profile) the torsional inertia may be calculated by the Extended Bredt-Bartho Shear Flow Theory (EBSFT) [66].

The process of calculating the shear center of a thin-walled multicellular closed section is composed of several steps. In order to calculate the shear center of these sections one has to: (a) produce an initially opened cross-section profile through a series of “cuts” equal to the number of closed cells; (b) apply an arbitrary shear force in order to calculate the shear stresses distribution associated with the opened profile; (c) use the null twist condition (shear center) in order to recover the shear stresses distribution at the closed section; and finally, (d) calculate the shear center coordinates. A detailed description about the shear center calculation of open and closed cross-sections is found in [49,67,68].

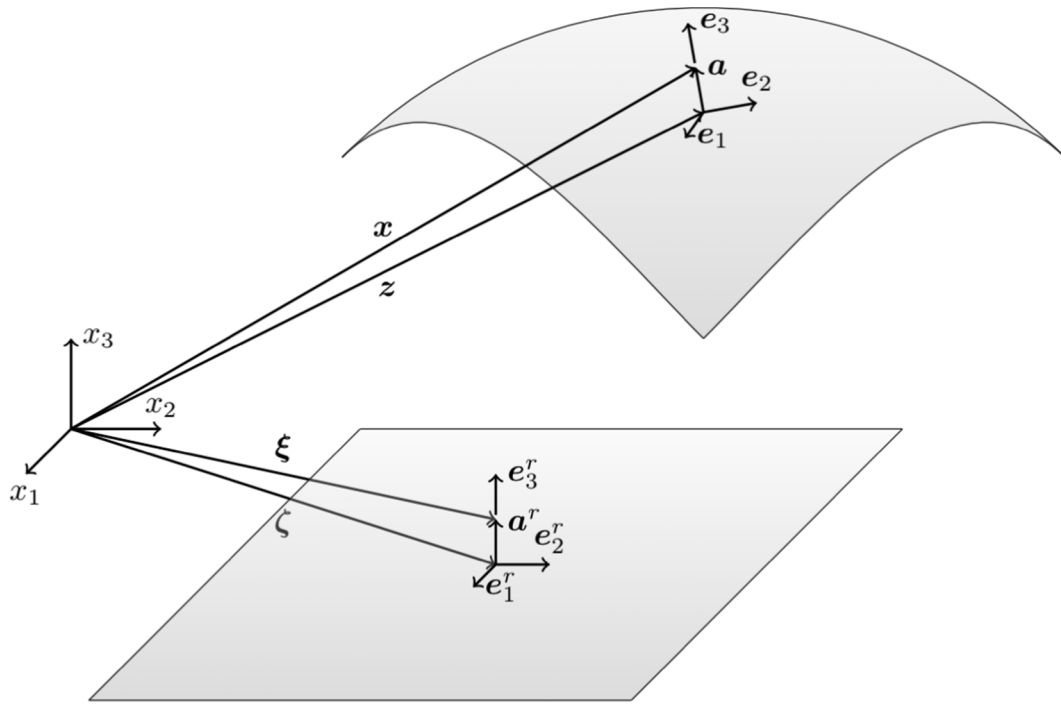


Fig. 3. Kinematic description of the Mindlin-Reissner shell model adopted.

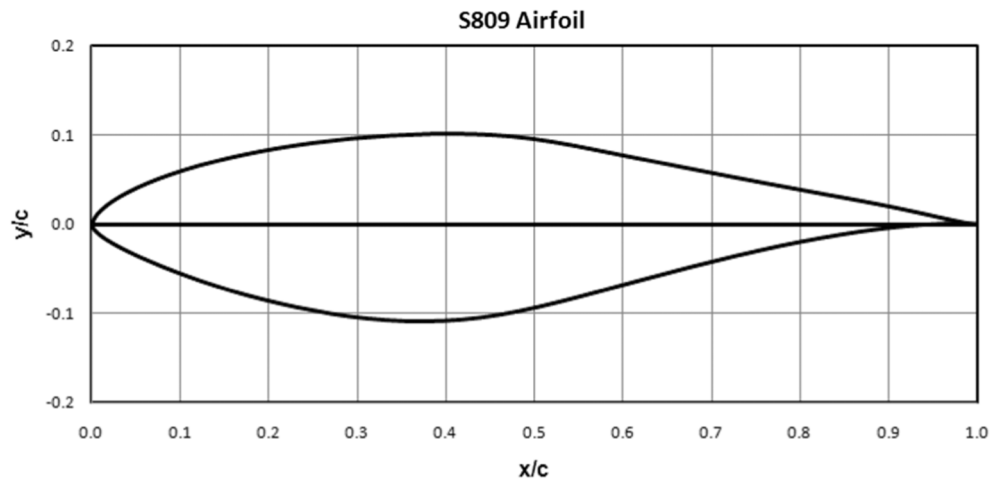


Fig. 4. S809 airfoil shape, extracted from [73].

4.2. Giraffe: a nonlinear finite element solver

The Generic Interface Readily Accessible For Finite Elements (*Giraffe*) is a finite element solver developed using C++ language, with the main objective of being a platform for the implementation of computational models, such as beam and shell finite element formulations, contact models, and others features [69]. *Giraffe* solver has been used in robust simulations of complex engineering problems such as oil exploitation risers [32,33,48], mechanisms [37], woven fabrics [70], wheel-rail contact [71], creased shells [54] and others.

Regarding the proposed comparative approach in the present work, *Giraffe* solver may be regarded as a suitable tool for the implementation of the presented models, once it presents both (beam and shell)

geometrically-exact formulations. Additionally, as presented in [72], it is possible to use the *Giraffe* solver in a full wind turbine dynamic simulation, including tower, nacelle, and blades. There are undergoing developments, for future works, with the objective of increasing the robustness of the wind turbine model, including aerodynamic loads and more elaborated geometric modeling.

5. Wind turbine blade model description

In order to approximate the proposed computational models to real wind turbine blades features, several realistic data are adopted. The data adopted regarding the proposed wind turbine blades are described in the sequence.



Fig. 5. S809 airfoil section including webs in green. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Table 1

Airfoil section thickness of each S809 airfoil part.

S809	Thickness (m)
Black parts	0.0184
Red parts	0.0581
Green parts	0.03975

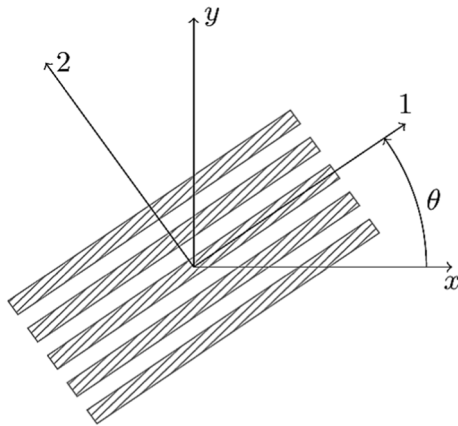


Fig. 6. Principal fibers coordinates positively oriented by an angle θ .

5.1. Airfoil section

The S809 wind turbine airfoil section was adopted [73]. According to [74], the S809 airfoil section was developed with the objective of achieving maximum lift coefficients (insensitive to roughness) and low

drag coefficients under certain conditions. The normalized chord coordinates “ x/c ” and “ y/c ” that define the S809 airfoil section may be found in [73,74]. It is possible, then, to plot the airfoil shape as presented in Fig. 4.

Once the airfoil shape is defined, it is necessary to define the cross-section webs. Webs are structural parts normally used to connect the upper surface to the lower surface of an airfoil. In fact, the webs positioning inside an airfoil is crucial for the overall wind turbine blade structural behavior. The webs not only increase the airfoil section stiffness but also prevent the local buckling of the section. In the proposed blade airfoil cross-section, two webs are adopted, based in the sample input file from the PreComp software [75]. Therefore, for the proposed model, the webs are defined as vertical structural parts approximately at 15% and 50% of the chord length. Fig. 5 presents the S809 airfoil cross-section including the webs.

Furthermore, it is necessary to define the thickness of each airfoil cross-section part. As for the webs, the thicknesses of the PreComp airfoil sample input files were taken as reference. Table 1 summarizes the airfoil thickness of each airfoil section part, according to the colors of Fig. 5.

5.2. Equivalent material

The structural part of modern HAWT blades is often composed of a stack of laminas with distinct material properties. Additionally, even considering that each lamina is made of an orthotropic material, the load may not be aligned with material principal directions. It is possible, however, to use the Classical Laminate Theory (CLT) to calculate equivalent material properties of a laminate composed of orthotropic laminas. According to [76], the CLT consists of a collection of hypotheses that permit the reduction of a 3D complex problem into a 2D solvable problem. Furthermore, the CLT assumes, for the laminate, the Kirchhoff hypothesis for plates [52] and the Kirchhoff-Love hypothesis for shells [55].

Table 2

Lamina information used for the calculation of the S809 equivalent material properties [75].

Name	t (m)	θ (deg.)	E_1 (GPa)	E_2 (GPa)	G_{12} (GPa)	ν_{12}	ρ (kg/m ³)
Unidirectional FRP	0.03975	30	37	9	4	0.28	1860
Double-bias FRP	0.01749	20	10.3	10.3	8	0.3	1830
Gelcoat	0.000381	0	1E–8	1E–8	1E–9	0.3	1830
Nexus	0.00051	0	10.3	1E+10	8	0.3	1664

Table 3
Equivalent properties for the S809 airfoil section.

E_{eq} (GPa)	14.0
G_{eq} (GPa)	5.6
ρ_{eq} (kg/m ³)	1849.06

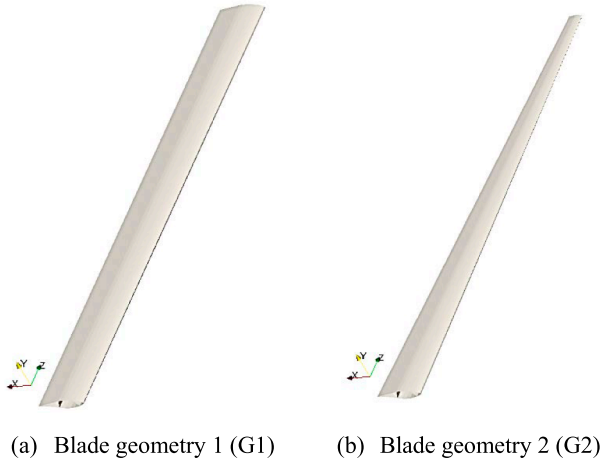


Fig. 7. Proposed blade geometries based on S809 airfoil.

Table 4
Proposed blade geometries description.

Blade geometry	Airfoil	Root/tip chord (m)	Reduction base point
G1	S809	2/2	–
G2		2/0.6	Geometric center

Thus, according to [76], the equivalent material properties E_x and G_{xy} of an orthotropic lamina positively oriented by an angle θ as presented in Fig. 6 are defined as

$$\frac{1}{E_x} = \frac{1}{E_1} \cos^4 \theta + \left(\frac{1}{G_{12}} - \frac{2\nu_{12}}{E_1} \right) \sin^2 \theta \cos^2 \theta + \frac{1}{E_2} \sin^4 \theta \quad \text{and} \quad (50)$$

$$\frac{1}{G_{xy}} = 2 \left(\frac{2}{E_1} + \frac{2}{E_2} + \frac{4\nu_{12}}{E_1} - \frac{1}{G_{12}} \right) \sin^2 \theta \cos^2 \theta + \frac{1}{G_{12}} (\sin^4 \theta + \cos^4 \theta) \quad (51)$$

where E_1 , E_2 , G_{12} and ν_{12} are the corresponding material properties associated with the fibers' principal directions.

Once the equivalent material properties of each lamina are calculated, the laminate equivalent properties E_{eq} and G_{eq} are simply defined as

$$E_{eq} = \frac{\sum_{i=1}^n E_{x,i} t_i}{\sum_{i=1}^n t_i} \quad \text{and} \quad (52)$$

$$G_{eq} = \frac{\sum_{i=1}^n G_{xy,i} t_i}{\sum_{i=1}^n t_i} \quad (53)$$

where the i iterator represents a lamina, n is the number of laminas and t_i is the thickness of each lamina. Additionally, it is also convenient to

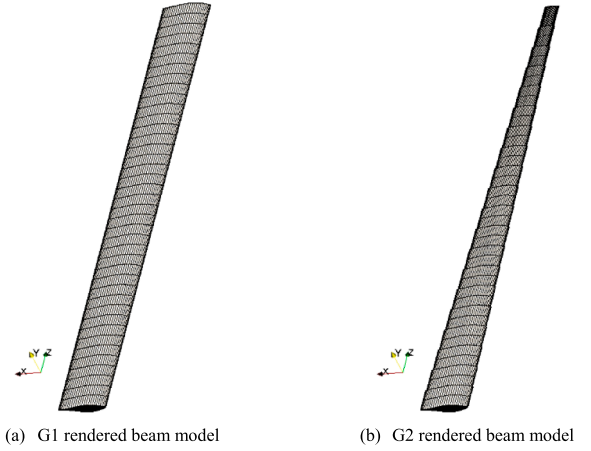


Fig. 8. Beam models discretization.

define an equivalent density of a laminate ρ_{eq} as

$$\rho_{eq} = \frac{\sum_{i=1}^n \rho_i t_i}{\sum_{i=1}^n t_i} \quad (54)$$

where the i iterator follows the same pattern described above and ρ_i is the density of each lamina.

Despite the beam formulation that incorporates only one isotropic homogeneous material, it is possible now to estimate equivalent material properties from a real laminate. For the proposed S809 airfoil cross-section material a laminate from the PreComp sample input file was taken as reference. Table 2 presents all lamina information used to calculate the equivalent material properties for the S809 airfoil.

Finally, with information from Table 2 and the previous equivalent material formulation, it is possible to calculate the resulting equivalent properties presented in Table 3.

5.3. Blade spam geometry

Despite the spanwise complex geometries that a wind turbine blade may present, in this article two simplified geometries based in the S809 airfoil are adopted. The first and simplest one (G1) is defined as a 20 m extrusion of an S809 airfoil with a 2 m chord length, thicknesses as presented in Table 1, and material as presented in Table 3. The second and more realistic one (G2) follows the same pattern of the G1 model; however, with a linear reduction in the S809 chord along length. In the G2 model, the airfoil chord reduces from 2 m at the root to 0.6 m at the tip, taking as a base point each cross-section geometric center. Fig. 7(a) presents the G1 model and Fig. 7(b) presents the G2 model. Table 4 summarizes the geometric features of the proposed blades.

It is important to mention that, regarding the formulations previously presented in this article, there is no limitation for the blade cross-section geometries. This feature is especially relevant in the context of modern HAWT blades.

5.4. Computational models

The G1 and G2 beam models are discretized into 20 3-node uniform meshes, with 1 m long elements. Fig. 8(a) and (b) present the rendered meshes for the beam models G1 and G2 respectively. This discretization proved to be adequate for the results presented here.

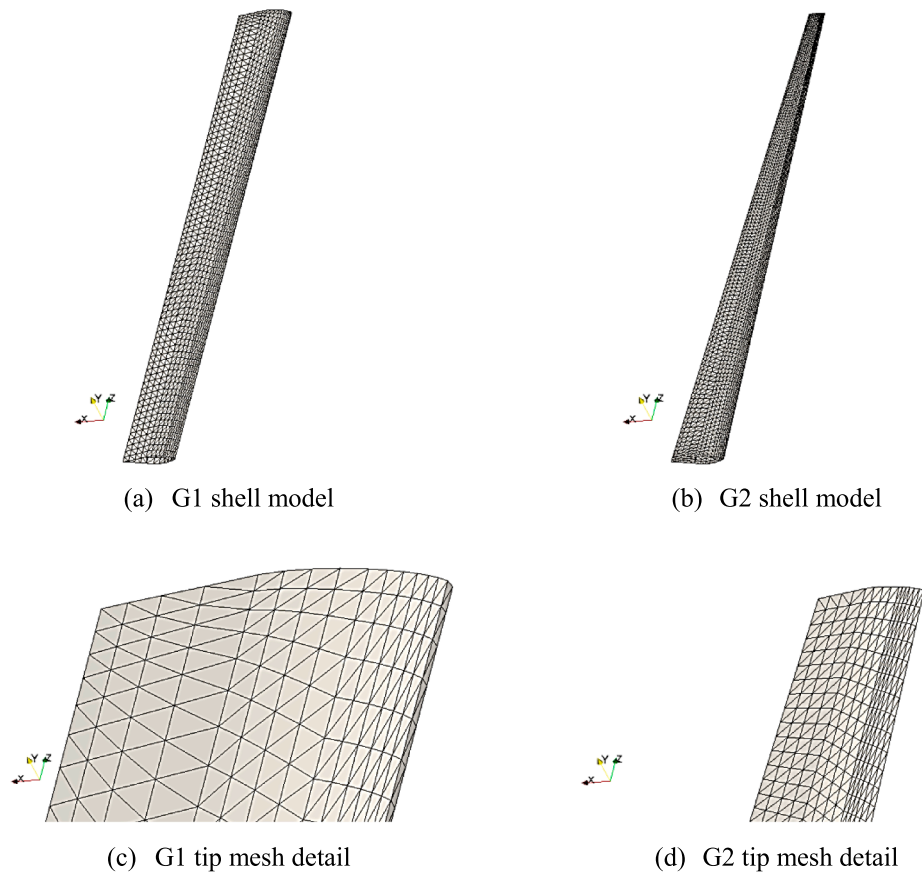


Fig. 9. Shell models discretization.

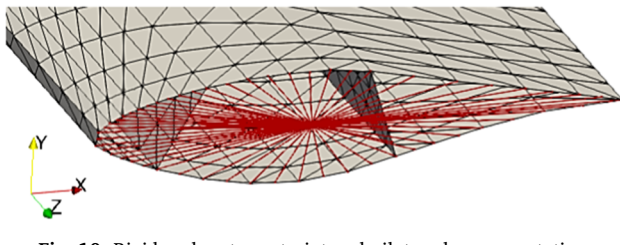


Fig. 10. Rigid node set constraint and pilot node representation.

The G1 shell model is discretized into 4884 6-node elements and the G2 shell model into 8976 6-node elements. Fig. 9(a) and (c) present the G1 shell model discretization, while Fig. 9(b) and (d) present the G2 shell model discretization.

6. Simulations and discussions

6.1. Proposed simulations

In order to compare beam and shell models, a series of static and dynamic load cases are proposed. For the static simulations, all models consider that root sections are clamped while pointwise external loads

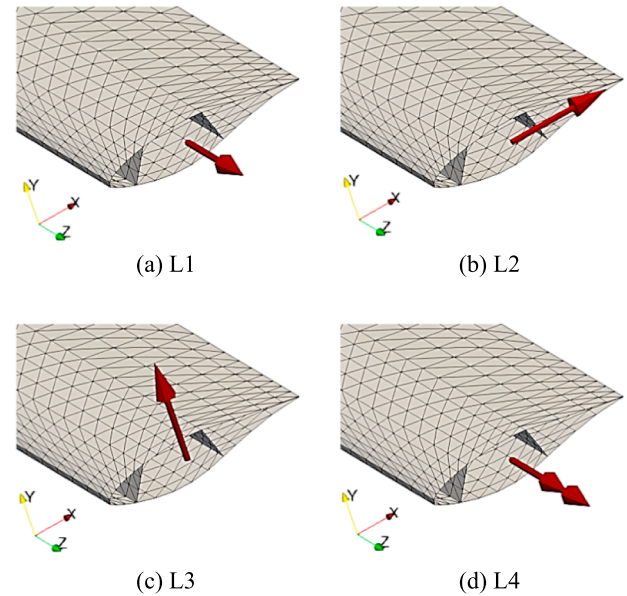


Fig. 11. Static simulations load cases representation.

Table 5

Proposed static simulations and analysis.

Load Case	Simulation	Global Direction	Load	Magnitude G1	Magnitude G2	Analysis
L1	Static	z	Force (N)	2.0E+8	1.4E+8	Displacements x, y and z
L2		x	Force (N)	3.5E+5	2.8E+5	Displacements x, y, z and rotation z
L3		y	Force (N)	2.0E+5	1.4E+5	Displacements x, y, z and rotation z
L4		z	Moment (N m)	6.0E+6	1.5E+6	Rotation z

Table 6
Natural periods and natural frequencies for the proposed blade geometries.

	G1				G2			
	Beam		Shell		Beam		Shell	
Mode	T (s)	f (Hz)	T (s)	f (Hz)	T (s)	f (Hz)	T (s)	f (Hz)
1	1.67	0.60	1.67	0.60	1.40	0.72	1.43	0.70
2	0.55	1.81	0.55	1.80	0.46	2.15	0.47	2.11
3	0.27	3.74	0.27	3.70	0.35	2.85	0.35	2.86
4	0.10	10.40	0.10	10.11	0.14	6.90	0.14	6.94
5	0.09	10.92	0.09	10.92	0.12	8.49	0.12	8.49
6	0.09	11.03	0.09	11.19	0.08	12.88	0.08	12.88
7	0.05	20.23	0.05	19.08	0.05	19.63	0.05	19.82
8	0.03	29.71	0.03	29.11	0.05	20.21	0.05	20.33
9	0.03	32.38	0.03	29.68	0.05	20.8	0.05	20.57
10	0.03	33.44	0.03	31.64	0.03	30.56	0.03	29.72
11	0.03	34.40	0.03	34.42	0.03	37.00	0.03	37.07
12	0.02	48.63	0.02	40.22	0.03	38.04	0.03	38.38

are applied at the tip of the blade, at the geometric center point calculated through *WindTurbine* tool. As the beam model axis corresponds to the locus of the cross-sections geometric centers, the loads are simply applied at the tip node of such models. For the shell model, a rigid node

set constraint, see [77], is defined, including all nodes at the blade tip cross-section and a node located at geometric center (pilot node). The rigid node set constraint and the pilot node are represented in Fig. 10. The objective of this constraint is to uniformly distribute the pointwise load applied at the pilot node.

The proposed load cases try to represent scenarios of extreme tension, bending, and torsion, even unrealistic examples, but that leads the models to their limits of representation of reality. The results of interest are the pilot node displacement/rotation responses for each case. Furthermore, we are interested in visualization of the deformed structure so as to see how each model captured local deformation patterns. Comparative plots are made to find out the main differences between beam and shell models. Table 5 summarizes the proposed static simulation load cases and Fig. 11 represents each simulation load, at the pilot node.

For the dynamic simulations, the only load presented for the structures is a prescribed displacement time-series with restrained rotations at the root cross-sections of the blade models. The displacement time-series defines an “8” shape circuit in order to induce excitation of the blade in all directions. Taking the middle point of the “8” circuit as an origin point, the displacement starts going in the direction of the Cartesian 1st quadrant and follows to the 2nd quadrant, 4th quadrant, and 3rd quadrant sequentially.

Table 7
Proposed dynamic simulations and analysis.

Load Case	Simulation	Prescribed Path	Circuit Time G1	Circuit Time G2	Analysis
L5	Dynamic	Symmetric “8” 1 m Diameter	1.7 s	1.5 s	Displacements x and y along time

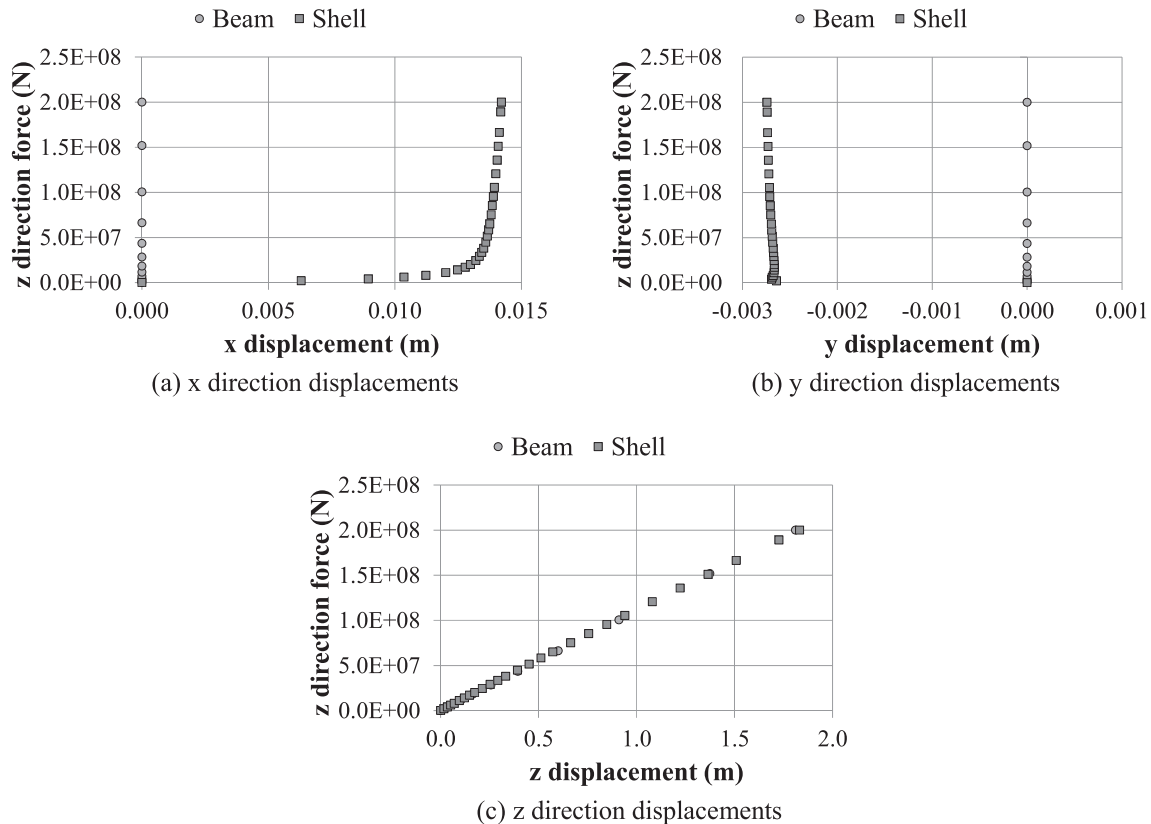


Fig. 12. L1 load case and G1 model results.

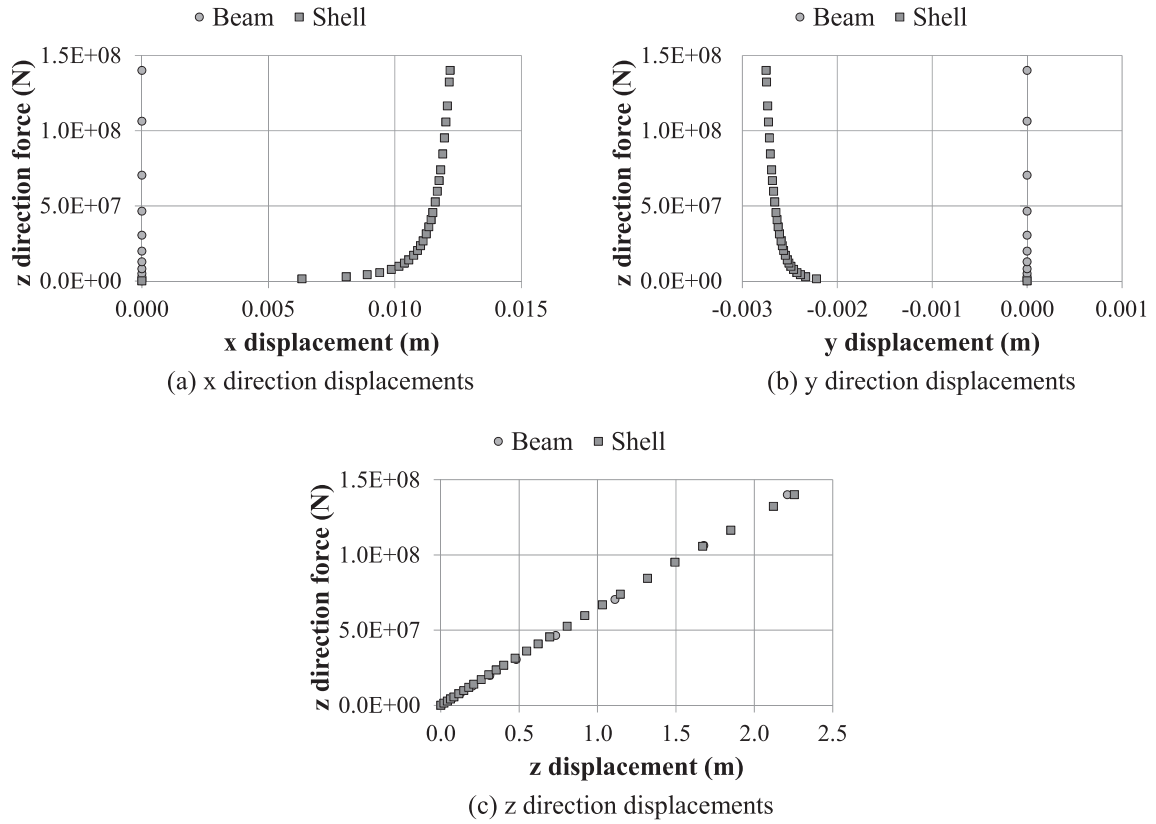


Fig. 13. L1 load case and G2 geometry results.

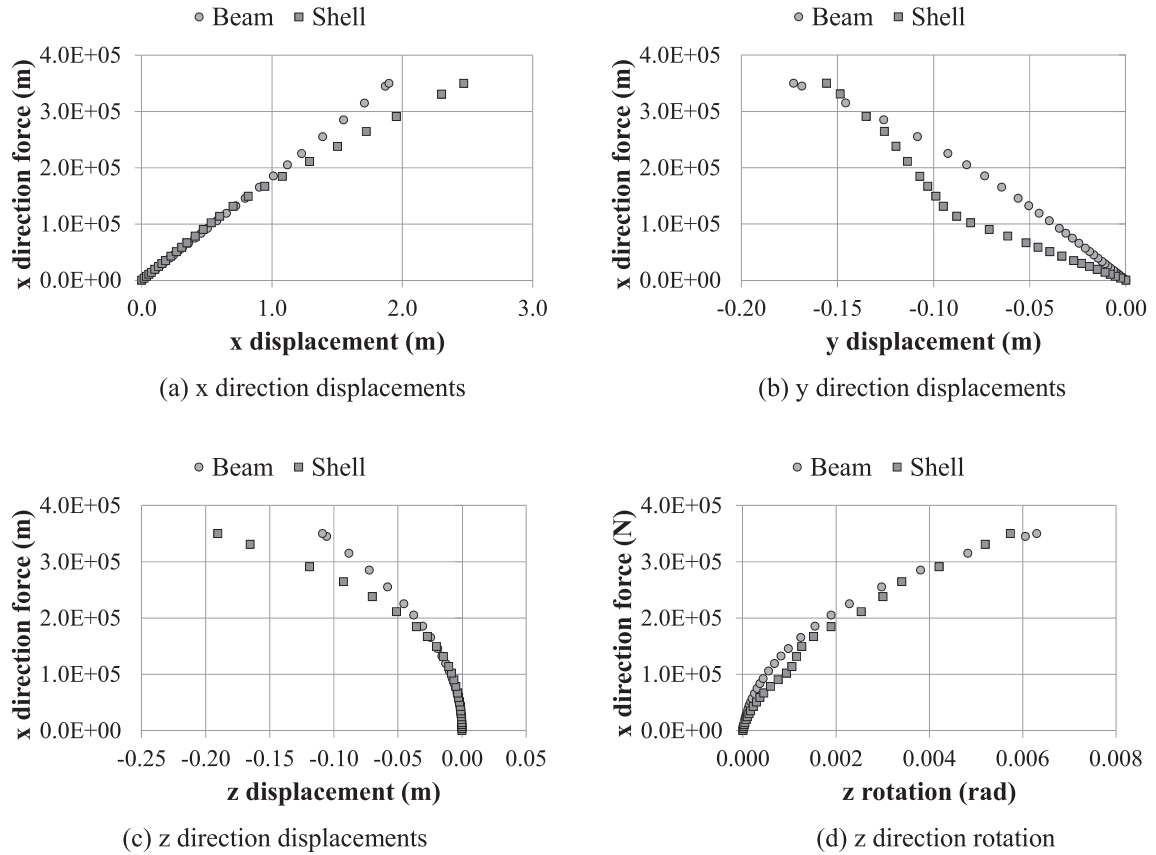


Fig. 14. L2 load case and G1 geometry results.

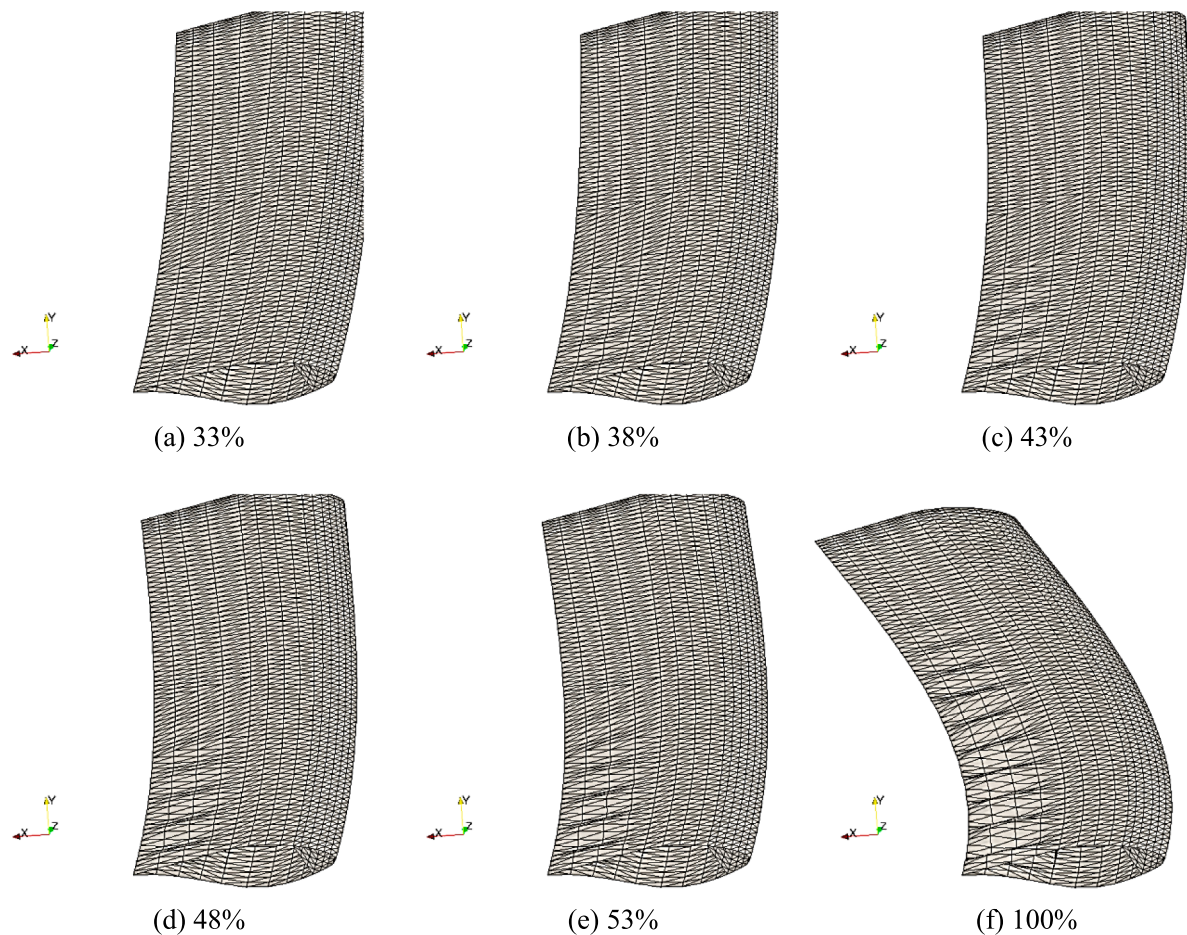


Fig. 15. Shell model local buckling evolution according to the end load percentage (G1L2).

The “8” shape circuit presents equal top and bottom parts with a diameter of 1 m. It is constructed by linear interpolated curves between points with a total circuit time of 1.7 s for the G1 model and of 1.5 s for the G2 model. The circuit time of each blade geometry was chosen based on the natural frequencies of the models as presented in Table 6 in order to ensure relevant dynamic contributions on system response.

It is important to mention that the dynamic simulations use Newmark integration method with parameters $\beta = 0.3$ and $\gamma = 0.5$, according to the formulation presented in [40] (this set of parameters induce negligible numerical damping). The time-step adopted is variable (adaptive) with an initial reference value of 0.01 s (also maximum) and permitting smaller values, according to difficulties faced during the nonlinear model solution. Tests employing smaller reference time-steps were done, leading to unimportant changes in overall kinematics.

The rigid node set constraint defined for the static simulations at tip cross-section of shell models is also kept at the dynamic simulations. Therefore, the structural analyses are also made with the aid of the pilot node responses and structure visualizations. Table 7 summarizes the proposed dynamic simulation.

7. Results and discussions

7.1. Load case 1 – static tension

The results of the L1 load case and the G1 model are presented in Fig. 12. It is seen in Fig. 12(a) and (b) that the x and y displacements present a similar pattern, but with a gap between the behaviors of the

beam and the shell models. The reason for this gap is the distinct location of the tension centers in both models. In the beam model, the tension center lays exactly by construction at the tip node, resulting in no bending due to tension load. However, in the shell model, the tension load occurs at the pilot node defined previously as the beam model tip node, calculated by the *WindTurbine* tool. Then, imprecisions due to the geometric characterization of the shells (e.g., mesh refinement) or due to simplifications assumed by the *WindTurbine* tool may lead to slight deviations from the beam models' results. However, numerically speaking, such deviations are really small when compared to the length of the blade (maximum 0.015 m displacement in 20 m length). Note that such deviations may be understood as bending of the blade, due to application of tension load slightly out of the tension center.

However, Fig. 12(c) shows that there is a very good concordance between beam and shell models, even for a large relatively large axial displacement.

The L1 results for the G2 model are presented in Fig. 13. The results of Fig. 13(a) and (b) present respectively a pattern similar to the ones of Fig. 12(a) and (b), for the same previously commented reasons. Moreover, Fig. 13(c) shows that the beam and shell models present a very similar response at axial direction, even considering large displacements.

7.2. Load case 2 – static bending about Y axis

The results associated with the L2 load case and the G1 model are presented in Fig. 14. It is seen in Fig. 14(a) that the beam and shell models present a similar response until approximately $1.8\text{E}+5\text{ N}$

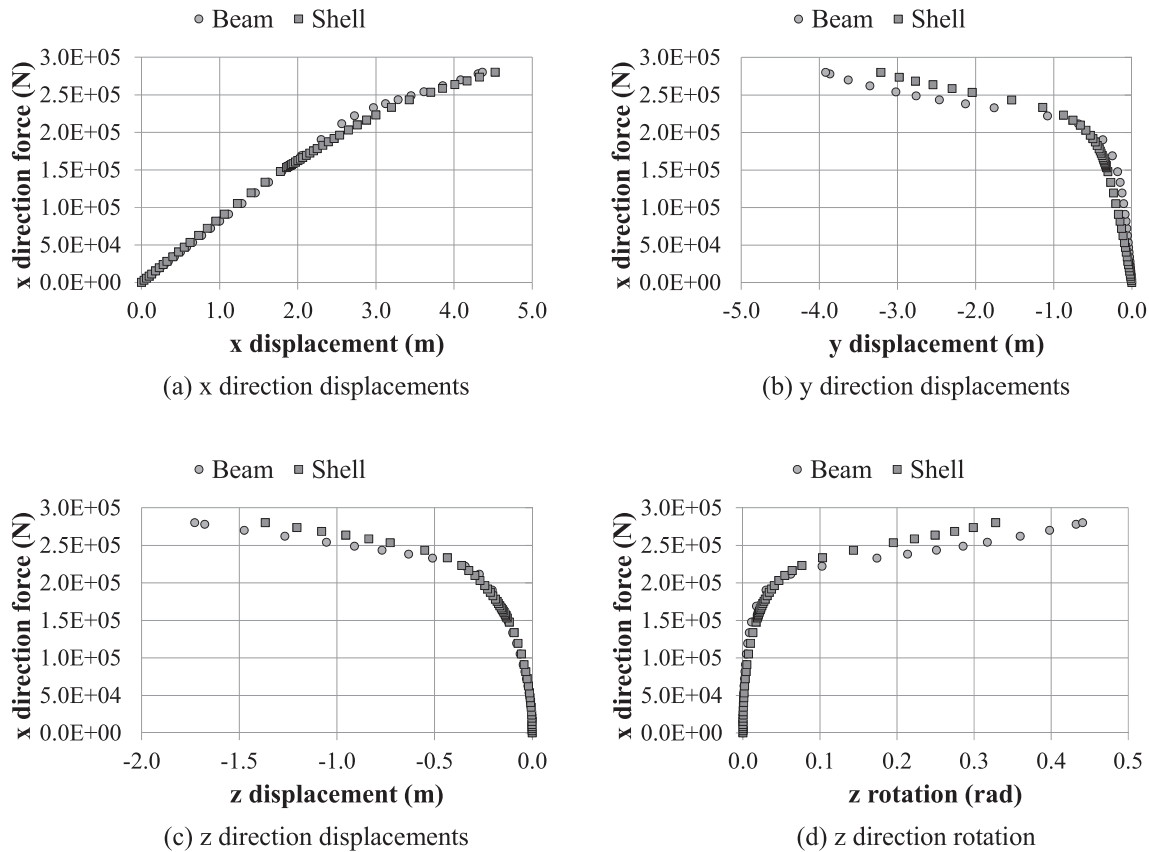


Fig. 16. L2 load case and G2 geometry results.

(approximately 47% of the end load). After this point, distinct overall stiffness responses occur. Additionally, the shell model behaves as if it is more flexible than the beam model for loads above 1.8×10^5 N. The analysis showed that the shell model presents local buckling occurrence close to the trailing edge, which is the reason for such deviation, since it may affect the whole model bending stiffness.

An interesting point is that even for loads below 1.8×10^5 N, local shell buckling can already be observed which means that for relatively low local buckling effects, the beam model may still provide a good overall response. In fact, the trailing edge local buckling has been a topic of concern for the wind energy industry due to the costs involved in repair and/or maintenance of the blades. In this context a series of structural solutions for the main wind turbine blades are seen in [16]. Fig. 15 presents the shell model local buckling evolution, according to the load percentage applied to the blade tip (pilot node).

Fig. 14(b) results show a quite different stiffness response between the beam and shell models. While the beam model presents a smooth softening nonlinear response, the shell model presents two apparently different responses with model stiffening at approximately 1.3×10^5 N (approximately 34% of the end load). The initial difference between the beam and shell models stiffness in this plot is likely to be due to the difference between the geometric centers and/or to the shear centers of both models, which may lead to distinct torsion-induced behavior. The distinct torsion behavior of the models may be also seen in Fig. 14(d), which shows the pilot node axial (z) rotation along load evolution. However, despite of the results of Fig. 14(a) that shows a major buckling effect starting approximately at 1.8×10^5 N, minor buckling effects may be also seen at 1.3×10^5 N as presented in Fig. 15.

In Fig. 14(c) it is seen that the beam and shell models present a good concordance until a load around 1.8×10^5 N (even considering minor buckling effects), from where the models present divergent responses. This response, as seen in Fig. 14(a), looks to be associated with the significantly local shell buckling of the structure.

The results of the L2 load case with respect to the G2 model are presented in Fig. 16. It is seen in Fig. 16(a) that the beam and shell models present a good concordance until approximately 2.0×10^5 N (approximately 71% of the end load). After that, the shell model stiffness decreases, indicating the same local buckling phenomena from previous analysis. In fact, as for the G1 model, the visualization of the shell model showed a local buckling at the trailing edge of the airfoil, just next to the root section. Fig. 17 presents the local buckling evolution, according to the end load percentage applied.

It is seen in Fig. 16(b) that both models present a good concordance, however divided into two distinct phases. At phase 1, when the loads are below 2×10^5 N, the beam and shell models present very similar responses with the beam model stiffness a bit higher than the shell model's. At phase 2, when the loads are above 2.0×10^5 N, both models' stiffness decreases and the shell model stiffness becomes higher than the beam model. The analyses of the results indicate that at phase 1, there is a predominant bending behavior, while at phase 2 there is a predominant tension behavior. Additionally, the stiffness difference between the models is likely to be associated with the local shell buckling that starts even at loads below 2.0×10^5 N as seen in Fig. 17.

The result in Fig. 16(c) indicates a very similar structural response between the models. As seen in the results of Fig. 16(b), the structural behavior may be also divided into two clear distinct phases by a load

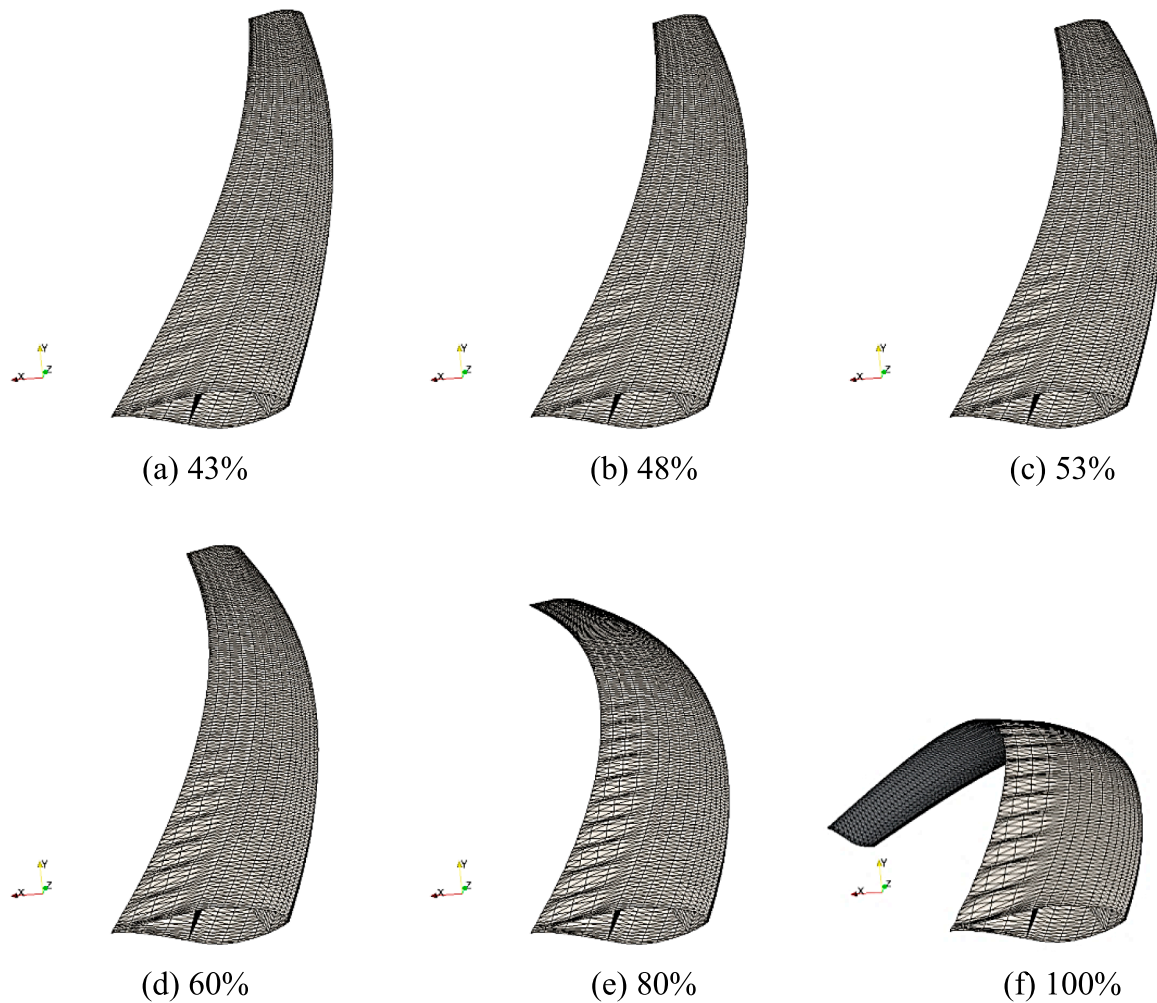


Fig. 17. Shell model local buckling evolution according to the end load percentage (G2L2).

around $2.0E+5$ N. At phase 1, the structural responses of both models present a good concordance with very similar stiffnesses. However, at phase 2, both models present slightly different structural responses, with the shell model stiffness being a bit higher than the beam model. The slightly different response of the models is likely to be associated with changes in the shell model stiffness caused by local buckling as seen in Fig. 17.

7.3. Load case 3 – static bending about X axis

Results concerning the load case L3 and G1 model are presented in Fig. 18. As seen in Fig. 18(a), the beam and shell models presented, at this direction, moderately different structural responses, due to distinct stiffness since the beginning of the simulations. The reason for that looks to be the same from the previous model, due to distinct positions of the shear centers and their consequences in the torsion response, as seen in Fig. 18(d). While the beam model shows an almost linear behavior, the shell model starts also with an almost linear behavior that changes to a nonlinear response approximately at $1.5E+5$ N, achieving an almost infinite stiffness. In fact, this effect is due to the low induced torsion of the shell model. This simulation analysis showed a small local buckling in the shell model near the root section as seen in Fig. 19. The local buckling and the difference between the geometric centers of the

models apparently played a major role at the small displacements associated with this simulation.

However, as seen in Fig. 18(b) and (c), the beam and shell models presented very similar responses at the y and z directions. It is reasonable to assume that the small local buckling of the shell model did not play a relevant role on the overall displacements since there is no abrupt change at the structural response. Moreover, it is important to note that the beam model could reproduce with a very good accuracy the shell model response even for the high nonlinear simulation imposed. Fig. 20 presents the overlapping of the rendered beam model in red and the shell model in gray at the end of the simulation.

The results for the L3 load case and G2 model are presented in Fig. 21. The result of Fig. 21(a) presents a similar response of the beam and shell models but with a stiffness gap between them. Once again, as seen in the previous simulations, this difference is likely to be associated with the difference in the induced torsion as seen in as seen in Fig. 21(d). In fact, this result enforces the hypothesis of the difference between the geometric centers and shear centers. However, unlike the result of Fig. 18(a), the shell model response presents no abrupt change. As expected, the analysis of the shell model, at this simulation, showed no local buckling.

The results of Fig. 21(b) and (c) present, as did the results of Fig. 18(b) and (c), a very good concordance between the beam and shell

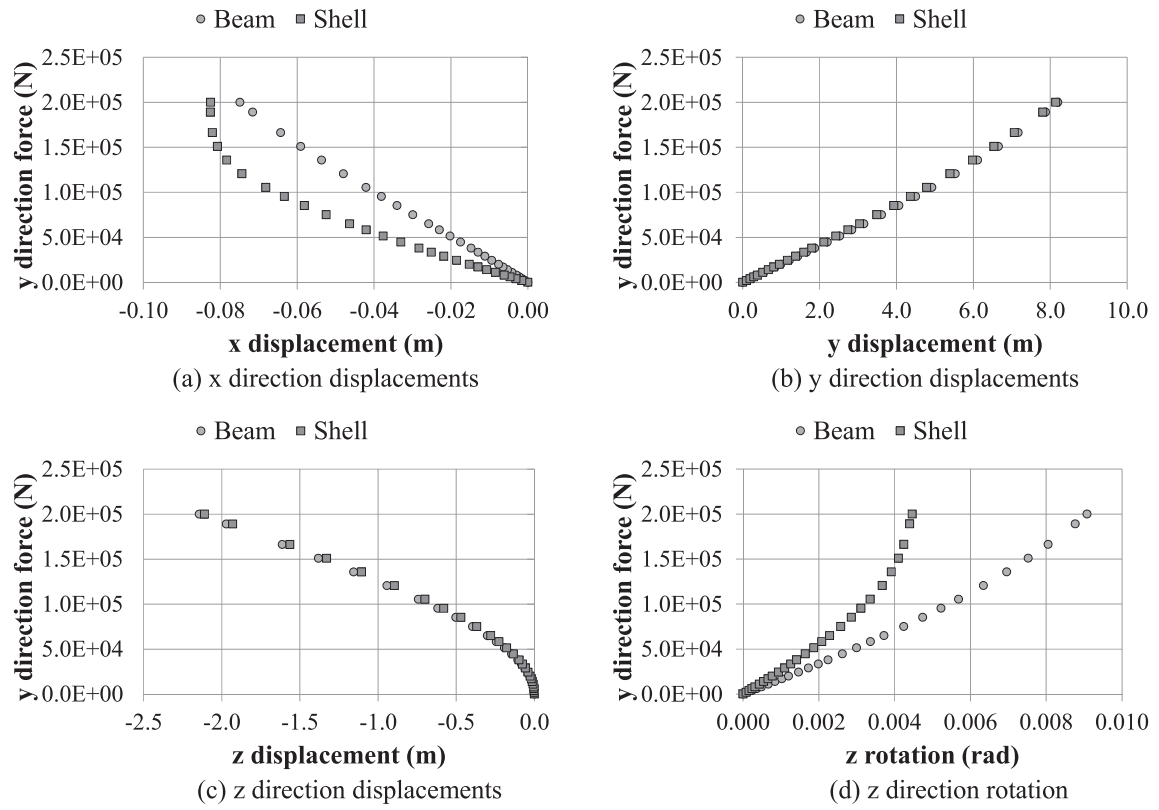


Fig. 18. L3 load case and G1 geometry results.

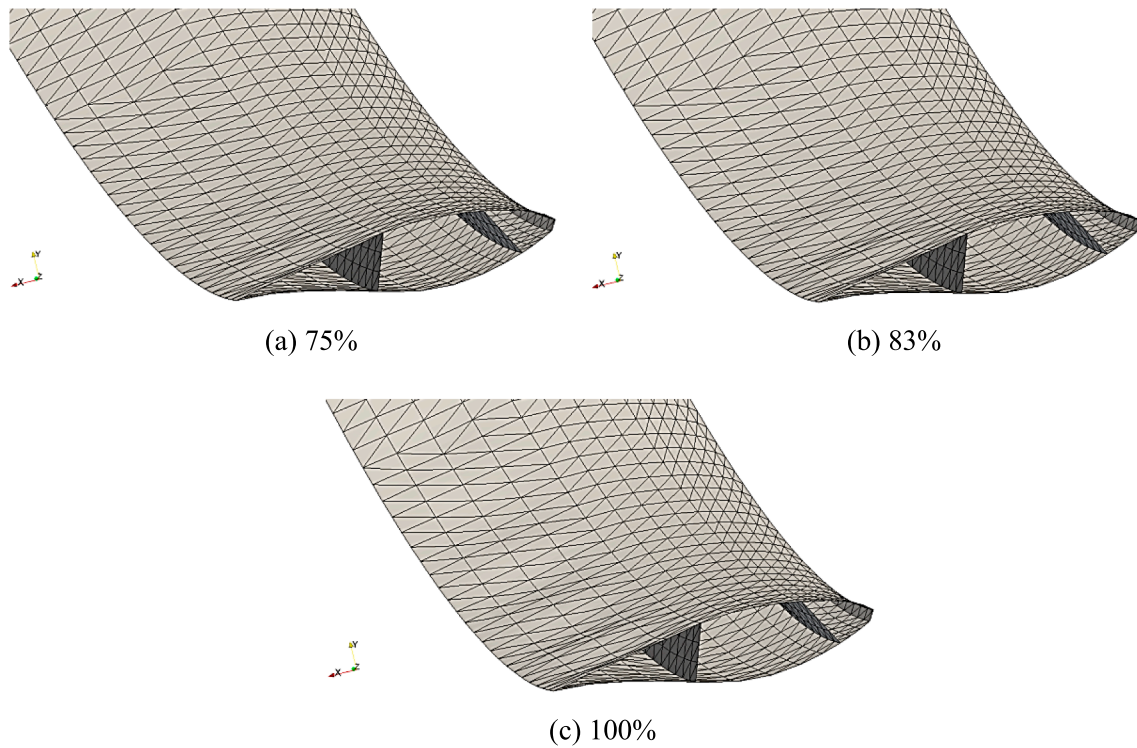


Fig. 19. Shell model local buckling evolution according to the end load percentage (G1L3).

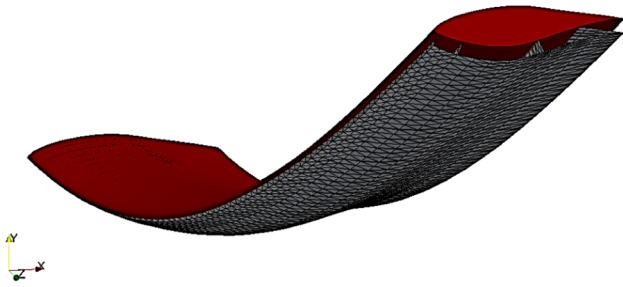


Fig. 20. Beam model in red and shell model in gray. Deformed shape at 100% load (G1L3). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

models, even for high nonlinear responses. The overlapping of the rendered beam model in red and the shell model in gray at the end of the simulation are seen in Fig. 22.

7.4. Load case 4 – static torsion

The result of the L4 load case with respect to the G1 model is presented in Fig. 23, which shows that beam and shell models presented very similar structural responses. However, while the beam model presented an almost linear response, the shell model presented a smooth nonlinear response. In fact, the visualization of the shell model response showed a significant buckling at the trailing edge along the whole blade. Therefore, it is reasonable to assume that this trailing edge buckling is highly likely to be associated with the shell model nonlinear response. Fig. 24 presents the buckling evolution at the trailing edge of the shell model according to the load percentage.



Fig. 22. Beam model in red and shell model in gray. Deformed shape at 100% load (G2L3). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

However, it is important to mention that regardless of the beam model assumption of rigid cross-section, the beam model could present a good accuracy with respect to the shell model, even considering the shell model local buckling. Moreover, the beam model torsion stiffness calculated through the thin-walled hypothesis at the *WindTurbine* tool showed to be very accurate in comparison with the shell model.

The result of the L4 load case with respect to the G2 model is presented in Fig. 25, which shows that both models presented an apparently linear structural response with a stiffness difference between them. In fact, at this simulation, the shell model visualization showed no local buckling.

However, the torsion stiffness of the models seems to be quite different since they rapidly diverge. Before the analyses of this result, it is

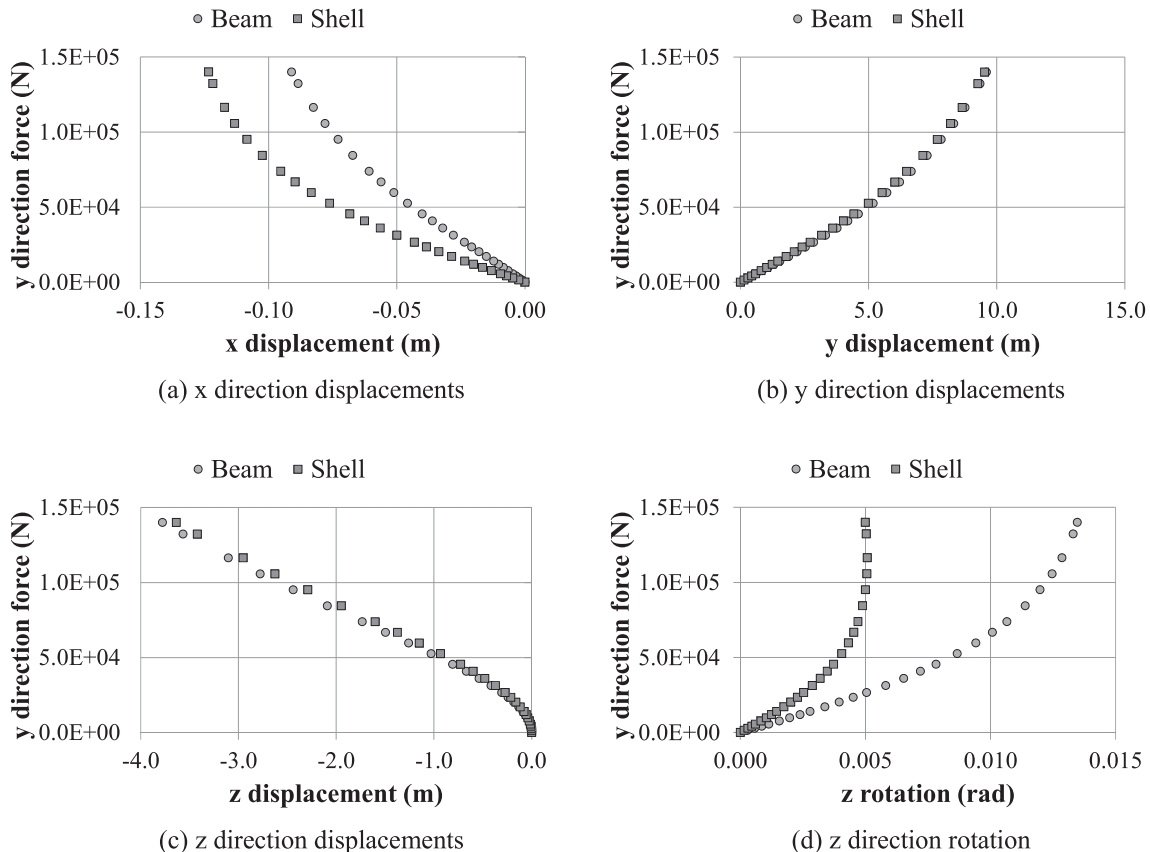


Fig. 21. L3 load case and G2 geometry results.

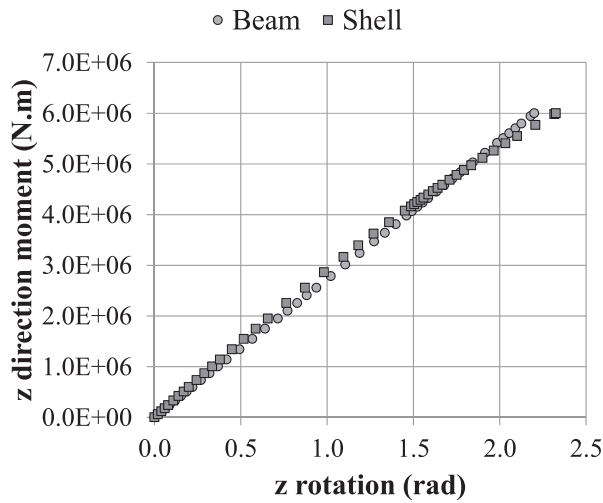


Fig. 23. L4 load case and G1 geometry results.

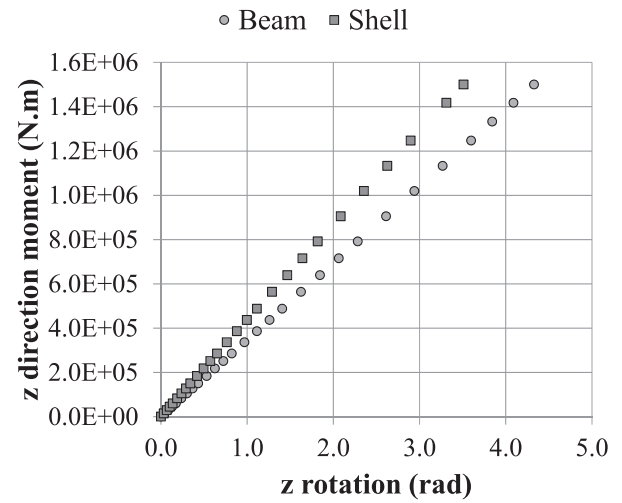


Fig. 25. L4 load case and G2 geometry results.

important to note that for the G2 model, the chord of the airfoil reduces from 2 m at the root section to 0.6 m at the tip section while the thickness of each airfoil part is kept. This condition leads, at the tip airfoil section, to a ratio of $\frac{t_{max}}{b} = \frac{0.058}{0.126} = 0.46$, which strictly does not respect the thin-walled hypothesis originally assumed for the *WindTurbine* tool. Moreover, as the chord reduces along the blade not only the airfoil tip section but also a substantial part of the blade does not respect the thin-walled hypothesis. Therefore, in this case the violation of the thin-walled hypothesis disqualifies the beam model results.

7.5. Load case 5 – dynamic “8” circuit

Results regarding the dynamic simulation for the G1 model are presented in Fig. 26. This Figure shows similar displacement shapes for

the beam and shell models along most of the simulation time. However, at some simulation time, the stiffness of the beam and shell model apparently diverge, being the shell model stiffness smaller than the beam model stiffness, especially at the horizontal displacements. The models non-concordance is mainly seen at the 1st and 3rd quadrants.

A more accurate analysis of the beam and shell model responses may be done through Fig. 27. Fig. 27(a) presents the structural response of the models along time for the x direction and Fig. 27(b) presents the structural response of the models along time for the y direction.

Fig. 27(a) shows that the beam and shell models have a similar structural response until approximately 0.95 s of simulation. From this point, the shell model displacement amplitude increases and virtually diverges from the beam model. Interestingly, the visualization of the shell model response showed a local buckling not only at 0.95 s, as

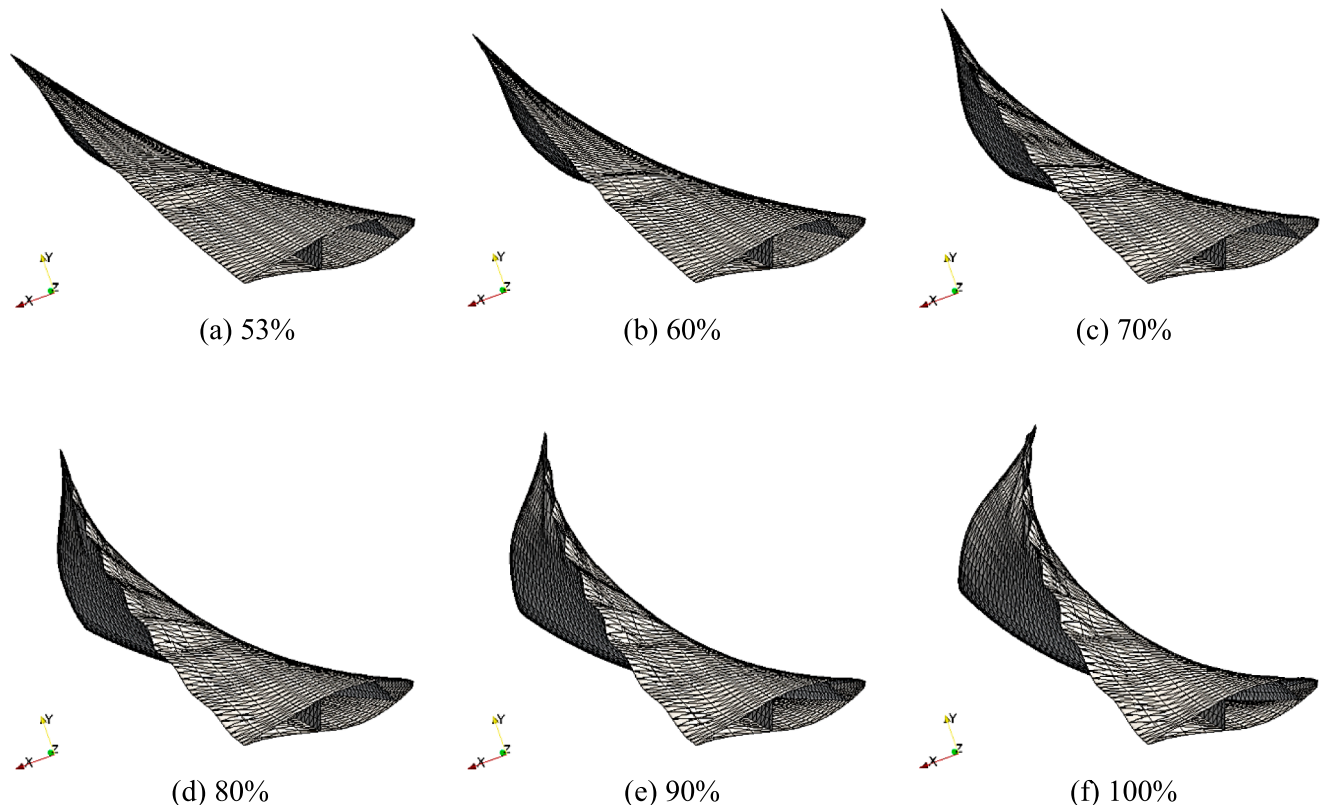


Fig. 24. Shell model local buckling evolution according to the end load percentage (G1L4).

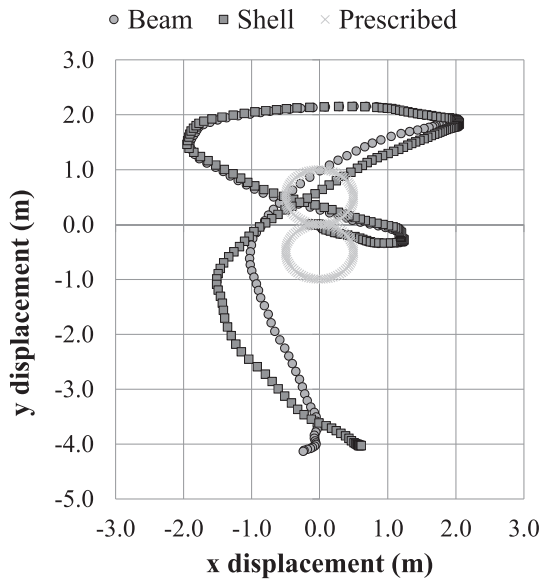


Fig. 26. L5 load case and G1 geometry blade tip displacement results. Prescribed displacements at the root cross-section.

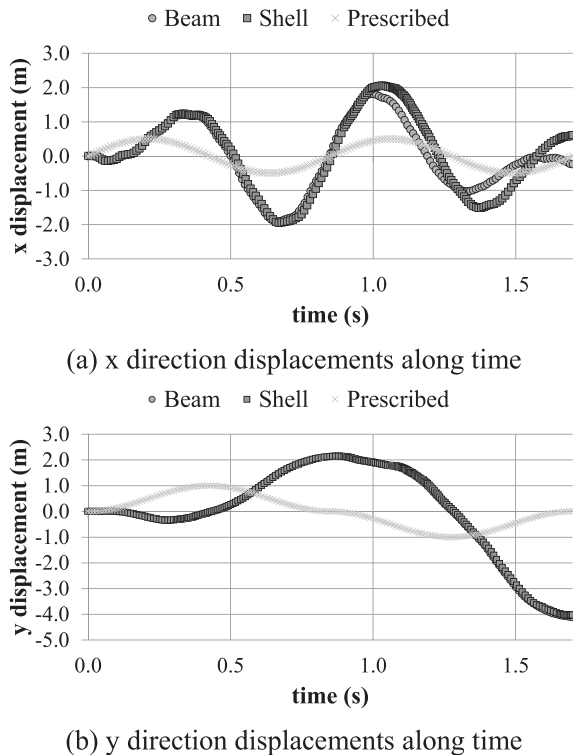


Fig. 27. Structural response along time for the L5 load case and G1 geometry blade tip displacement results. Prescribed displacements at the root cross-section.

expected, but also around 0.35 s with a small intensity. Fig. 28 presents the shell model local buckling evolution in both cases. Once again, the beam model has proven to have good global responses even neglecting such small buckling effects.

However, Fig. 27(b) shows that at y direction, the beam and shell

models present a very good accuracy. It is interesting to note that the local buckling of the shell model at approximately 0.95 s does not play a role in the structural response of the model in this direction. This response indicates that the shell model local buckling affects mainly the x direction global behavior.

The results regarding the dynamic simulation for the G2 model are presented in Fig. 29. This Figure shows that the beam and shell models have similar structural behaviors at most of the simulation time. However, as seen in the previous simulation, the shell model apparently loses stiffness at some simulation time when compared to the beam model stiffness.

However, the structural response of the models may be more precisely evaluated by the results of Fig. 30. Fig. 30(a) and (b) show the structural response of the beam and shell models with respect to the x displacements and y displacements respectively. Fig. 30(a) shows that the beam and shell models have a good concordance until approximately 0.85 s of simulation. From that point on, the shell model presents a substantially higher horizontal displacement and after a quite different structural response in comparison to the beam model. In fact, the visualization of the shell model response showed a significant local buckling not only at 0.85 s but also at 0.35 s as seen in Fig. 31. It is important to note that, as for the previous simulation, the beam model showed to represent the global behavior, even neglecting the small local buckling around 0.35 s. The beam model results clearly do not agree with the shell's when significant local buckling takes place at 0.85 s.

Furthermore, as seen in Fig. 30(b), the beam and shell models present very similar structural responses, even regarding the high amplitudes and the shell model's buckling effects involved. Therefore, it is possible to conclude that the shell model buckling affects more the x direction than y direction displacements.

8. Conclusions

This work presented a comparison between beam and shell models. Static and dynamic scenarios were considered, in which loads were chosen to induce relevant geometric nonlinearities, in order to test the limits of representation of the models. From the engineering point of view, the geometrically-exact beam model could capture very well the overall structural response when compared to shell models. High level of bending and torsion were induced and a good agreement between the models was achieved, including capturing nonlinear overall behavior.

The main drawback of the adopted beam model is the assumption of a rigid cross-section. It is not capable of capturing local buckling phenomena. This caused some deviations in comparative results with shells when local buckling takes place close to wind turbine blade trailing edge. With respect to global stiffness of the blade, the influence of such local buckling showed to affect more the bending stiffness of the blade about the larger inertia direction. Less influence of local buckling in bending stiffness about the smaller inertia direction was observed. It is remarkable to note that the beam models could be very well responsive even when small local buckling of the shell model occurs.

It is possible to state that geometrically-exact beam models may be seen as a reliable choice for overall system design since they produce fast and good-quality results. However, it is recommended always to perform shell models for selected candidates during design in order to verify local buckling occurrence, which is not captured by beam models and may affect local stresses in a dangerous way for wind turbine blades. In these cases, the global response can also be re-verified in such shell models.

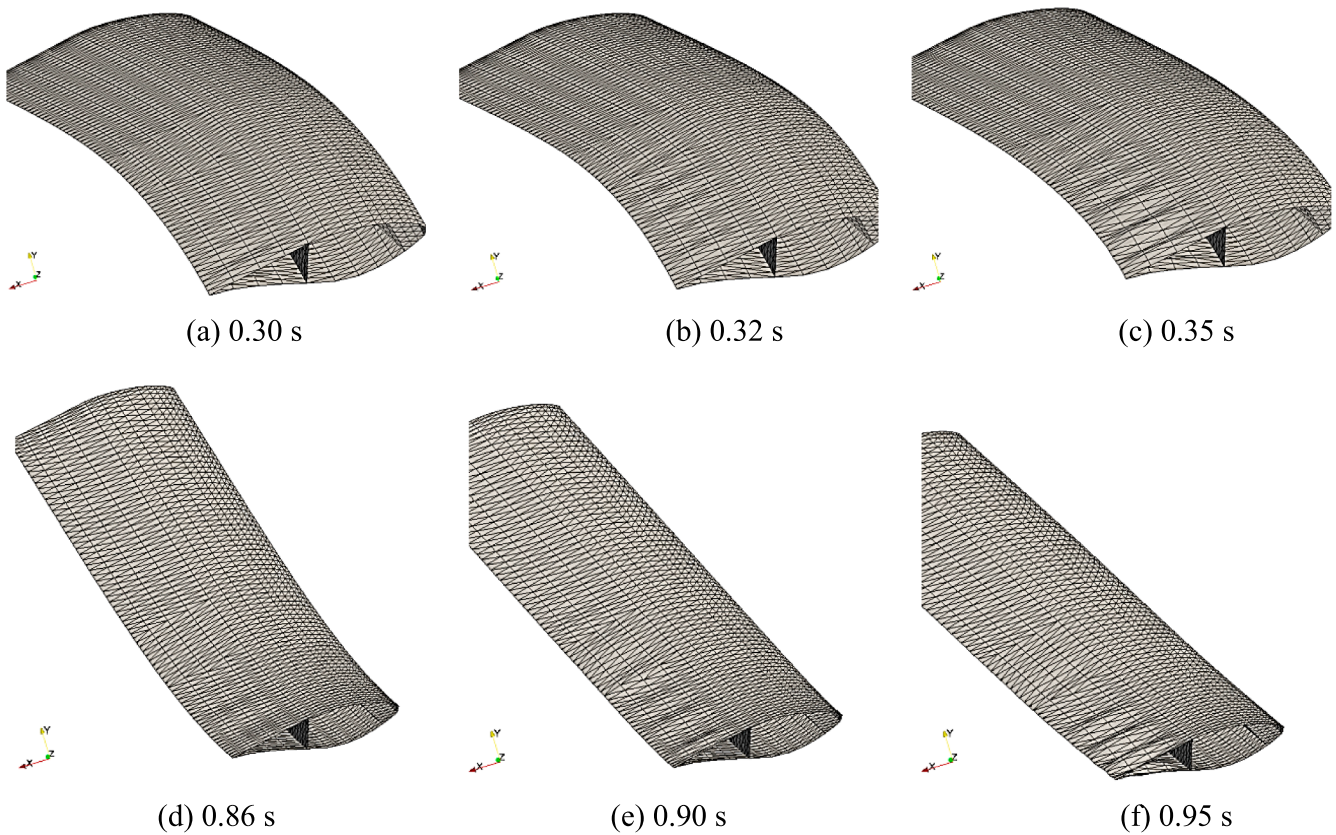


Fig. 28. Shell model buckling at dynamic simulation (G1L5).

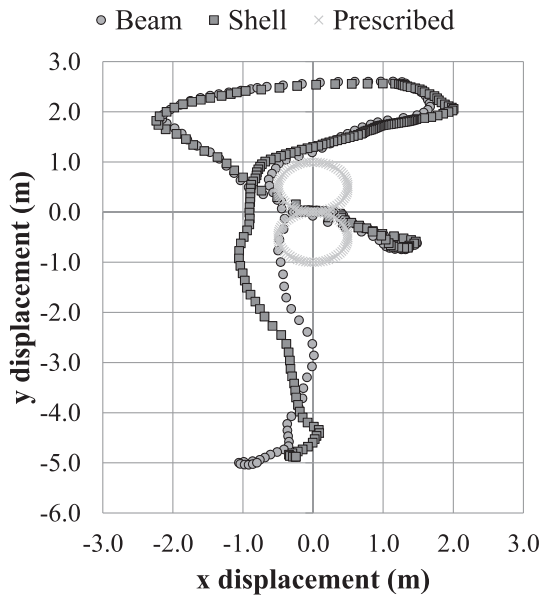
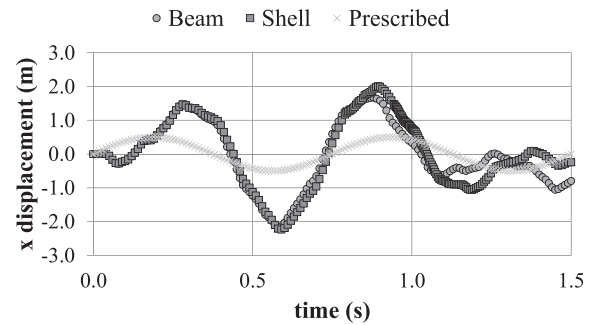
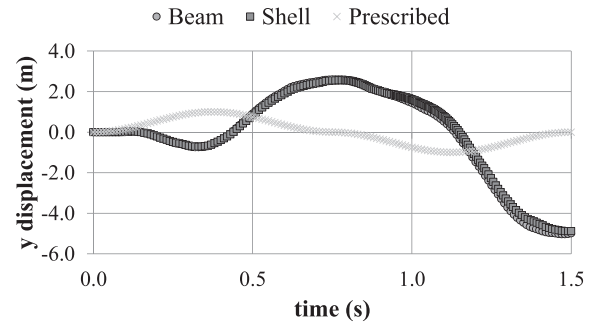


Fig. 29. L5 load case and G2 geometry blade tip displacement results. Prescribed displacements at the root cross-section.

In future work, a more complete beam and/or shell formulation could be adopted considering the composite materials actually used on modern wind turbine blades. Alternatively, one may adopt a more realistic and complex wind turbine blade geometry, including an initial curvature pattern for the blade reference axis shape.



(a) x direction displacements along time



(b) y direction displacements along time

Fig. 30. Structural response along time for the L5 load case and G2 geometry blade tip displacement results. Prescribed displacements at the root cross-section.

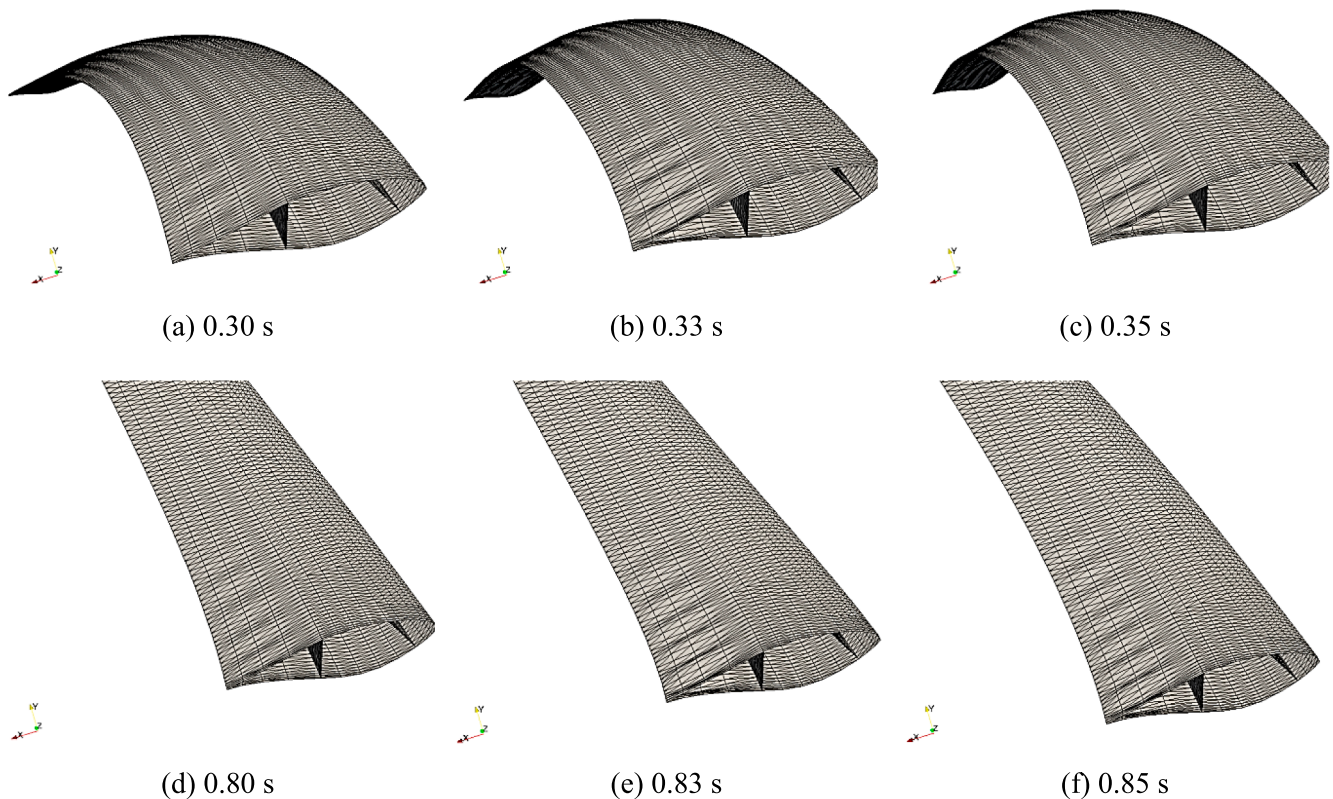


Fig. 31. Shell model buckling of the dynamic simulation (G2L5).

Acknowledgements

The authors acknowledge FAPESP (Fundação de Amparo à Pesquisa do Estado de São Paulo) for funding the project under grant number 2015/11655-3. Authors also acknowledge the Office of Naval Research/Global for funding part of this work through the Research Grant NICOP - Computational Tools for Research on Floating Wind Turbines (Award # N62909-16-1-2066), CNPq (Conselho Nacional de Desenvolvimento Científico e Tecnológico) under the grant 308190/2015-7 and CAPES (Coordenação de Aperfeiçoamento de Pessoal de Nível Superior).

Appendix A. Supplementary material

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.engstruct.2018.09.032>.

References

- [1] Wind Power Chalks Up More Strong Numbers; 2017. <http://gwec.net/wind-power-chalks-up-more-strong-numbers/> [accessed May 22, 2018].
- [2] Gardiner G, HAWTs vs. VAWTs; 2011. <https://www.compositesworld.com/articles/hawts-vs-vawts> [accessed May 22, 2018].
- [3] V.W.S. A/S, Vestas V164-8.0; 2015. <https://en.wind-turbine-models.com/turbines/318-vestas-v164-8.0> [accessed May 22, 2018].
- [4] Bak C, Zahle F, Bitsche R, Kim T, Yde A, Henriksen LC, et al. Reference wind turbine. Danish Wind Power Res 2013.
- [5] Berg JC, Resor BR. Numerical manufacturing and design tool (NuMAD v2. 0) for wind turbine blades: user's guide 2012, SAND2012-7028. n.d.
- [6] Verelst D. Flexible Wind turbine Blades: a BEM-FEM coupled model approach, Delft Univ. Technol. (TU Delft); 2009.
- [7] ABAQUS User's Manual - Version 6.5, Inc., Pawtucket, RI; 2004.
- [8] Moriarty PJ, Hansen AC. AeroDyn theory manual, Citeseer; 2005.
- [9] Yu DO, Kwon OJ. Predicting wind turbine blade loads and aeroelastic response using a coupled CFD-CSD method. Renew Energy 2014;70:184–96.
- [10] Hsu MC, Bazilevs Y. Fluid-structure interaction modeling of wind turbines: simulating the full machine. Comput Mech 2012;50:821–33. <https://doi.org/10.1007/s00466-012-0772-0>.
- [11] Hayat K, de Lecea AGM, Moriones CD, Ha SK. Flutter performance of bend-twist coupled large-scale wind turbine blades. J Sound Vib 2014;370:149–62. <https://doi.org/10.1016/j.jsv.2016.01.032>.
- [12] Wang L, Quant R, Kolios A. Fluid structure interaction modelling of horizontal-axis wind turbine blades based on CFD and FEA. J Wind Eng Ind Aerodyn 2016;158:11–25. <https://doi.org/10.1016/j.jweia.2016.09.006>.
- [13] ANSYS User's Manual, Inc. Model. CFX. 11; 2000.
- [14] Montesano J, Chu H, Singh CV. Development of a physics-based multi-scale progressive damage model for assessing the durability of wind turbine blades. Compos Struct 2016;141:50–62. <https://doi.org/10.1016/j.compstruct.2016.01.011>.
- [15] Wang L, Kolios A, Nishino T, Delafin PL, Bird T. Structural optimisation of vertical-axis wind turbine composite blades based on finite element analysis and genetic algorithm. Compos Struct 2016;153:123–38. <https://doi.org/10.1016/j.compstruct.2016.06.003>.
- [16] Bladerna, n.d. <https://www.bladerna.com/> [accessed March 4, 2018].
- [17] Wang L, Liu X, Kolios A. State of the art in the aeroelasticity of wind turbine blades: aeroelastic modelling. Renew Sustain Energy Rev 2016;64:195–210. <https://doi.org/10.1016/j.rser.2016.06.007>.
- [18] Hansen MOL, Sørensen JN, Voutsinas S, Sørensen N, Madsen HA. State of the art in wind turbine aerodynamics and aeroelasticity. Prog Aerosp Sci 2006;42:285–330. <https://doi.org/10.1016/j.paerosci.2006.10.002>.
- [19] Zhang P, Huang S. Review of aeroelasticity for wind turbine: current status, research focus and future perspectives. Front Energy 2011;5:419–34. <https://doi.org/10.1007/s11708-011-0166-6>.
- [20] Wang L, Wang T, Luo Y. Improved non-dominated sorting genetic algorithm (NSGA-II) in multi-objective optimization studies of wind turbine blades. Appl Math Mech 2011;32:739–48. <https://doi.org/10.1007/s10483-011-1453-x>.
- [21] Sessarego M, Dixon KR, Rival DE, Wood DH. A hybrid multi-objective evolutionary algorithm for wind-turbine blade optimization. Eng Optim 2015;47:1043–62. <https://doi.org/10.1080/0305215X.2014.941532>.
- [22] Larsen TJ, Hansen AM. How 2 HAWC2, the user's manual; 2015.
- [23] HAWC2 - Structural Formulation, 2015. <http://www.hawc2.dk/hawc2-info/structural-formulation> [accessed May 22, 2018].
- [24] Wang L, Liu X, Renevier N, Stables M, Hall GM. Nonlinear aeroelastic modelling for wind turbine blades based on blade element momentum theory and geometrically exact beam theory. Energy 2014;76:487–501. <https://doi.org/10.1016/j.energy.2014.08.046>.
- [25] Hodges DH, D. Ph. Nonlinear Beam Theory for Engineers Georgia Institute of Technology; 1994.
- [26] Wang Q, Johnson N, Sprague MA, Jonkman J. Beamdyn: a high-fidelity wind turbine blade solver in the FAST modular framework. In: 33rd Wind Energy Symp. 2015; 2015.
- [27] Gay Neto A. Wind turbine dynamics; 2017. <http://sites.poli.usp.br/p/alfredo.gay/models.html> [accessed May 22, 2018].
- [28] Gay Neto A. Giraffe; 2016. <http://sites.poli.usp.br/p/alfredo.gay/giraffe.html> [accessed May 22, 2018].

- [29] Gay Neto A. Giraffe User's Manual v. 1.0.218, 2017. http://sites.poli.usp.br/p/alfredo.gay/GIRAFFE_manual.pdf.
- [30] Bauchau OA, Bottasso CL, Nikishkov YG. Modeling rotorcraft dynamics with finite element multibody procedures. *Math Comput Model* 2001;33:1113–37. [https://doi.org/10.1016/S0895-7177\(00\)00303-4](https://doi.org/10.1016/S0895-7177(00)00303-4).
- [31] Gay Neto A, de Martins C de A. Structural stability of flexible lines in catenary configuration under torsion. *Mar Struct* 2013;34:16–40. <https://doi.org/10.1016/j.marstruc.2013.07.002>.
- [32] Gay Neto A, Martins CA, Pimenta PM. Static analysis of offshore risers with a geometrically-exact 3D beam model subjected to unilateral contact. *Comput Mech* 2014;53:125–45. <https://doi.org/10.1007/s00466-013-0897-9>.
- [33] Gay Neto A, Ribeiro Malta E, Pimenta PM. Catenary riser sliding and rolling on seabed during induced lateral movement. *Mar Struct* 2015;41:223–43. <https://doi.org/10.1016/j.marstruc.2015.02.001>.
- [34] E.M.B. Campello, Análise Não-Linear de Perfis Metálicos Conformados a Frio; 2000.
- [35] Campello EMB, Lago LB. Effect of higher order constitutive terms on the elastic buckling of thin-walled rods. *Thin-Walled Struct* 2014;77:8–16. <https://doi.org/10.1016/j.tws.2013.11.001>.
- [36] Manta D, Gonçalves R. A geometrically exact Kirchhoff beam model including torsion warping. *Comput Struct* 2016;177:192–203. <https://doi.org/10.1016/j.compstruc.2016.08.013>.
- [37] Gay Neto A. Simulation of mechanisms modeled by geometrically-exact beams using Rodrigues rotation parameters. *Comput Mech* 2016. <https://doi.org/10.1007/s00466-016-1355-2>.
- [38] Bauchau OA, Damilano G, Theron NJ. Numerical integration of nonlinear elastic dynamics using finite elements. *Comput Methods Appl Mech Eng* 1995;38:2727–51. <https://doi.org/10.1002/nme.1620381605>.
- [39] Cardona A, Geradin M, Doan DB. Rigid and flexible joint modelling in multibody dynamics using finite elements. *Comput Methods Appl Mech Eng* 1991;89:395–418. [https://doi.org/10.1016/0045-7825\(91\)90050-G](https://doi.org/10.1016/0045-7825(91)90050-G).
- [40] Ibrahimbegović A, Mamouri S. On rigid components and joint constraints in nonlinear dynamics of flexible multibody systems employing 3D geometrically exact beam model. *Comput Methods Appl Mech Eng* 2000;188:805–31. [https://doi.org/10.1016/S0045-7825\(99\)00363-1](https://doi.org/10.1016/S0045-7825(99)00363-1).
- [41] Simo JC. A finite strain beam formulation. The three-dimensional dynamic problem. Part I. *Comput Methods Appl Mech Eng* 1985;49:55–70. [https://doi.org/10.1016/0045-7825\(85\)90050-7](https://doi.org/10.1016/0045-7825(85)90050-7).
- [42] Reissner E. On one-dimensional finite-strain beam theory: the plane problem. *Zeitschrift für Angew Math Und Phys ZAMP* 1972;23:795–804. <https://doi.org/10.1007/BF01602645>.
- [43] Love AEH. A treatise on the mathematical theory of elasticity, 1; 1892.
- [44] Pimenta PM, Yojo T. Geometrically exact analysis of spatial frames. *Appl Mech Rev* 1993;46:S118–28. <https://doi.org/10.1115/1.3122626>.
- [45] Yojo T. Análise Não-Linear Geometricamente Exata de Pórticos Espaciais; 1993.
- [46] Simo JC, Vu-Quoc L. A Geometrically-exact rod model incorporating shear and torsion-warping deformation. *Int J Solids Struct* 1991;27:371–93. [https://doi.org/10.1016/0020-7683\(91\)90089-X](https://doi.org/10.1016/0020-7683(91)90089-X).
- [47] Pimenta PM, Campello EMB. Geometrically nonlinear analysis of thin-walled space frames; In: 2nd Eur congr comput mech; 2001.
- [48] Gay Neto A. Dynamics of offshore risers using a geometrically-exact beam model with hydrodynamic loads and contact with the seabed. *Eng Struct* 2016;125:438–54. <https://doi.org/10.1016/j.engstruct.2016.07.005>.
- [49] Paz M, Strehl CP, Schrader P. Computer determination of the shear center of open and closed sections. *Comput Struct* 1976;6:117–25. [https://doi.org/10.1016/0045-7949\(76\)90061-4](https://doi.org/10.1016/0045-7949(76)90061-4).
- [50] Megson THG. Introduction to aircraft structural analysis. 2nd ed. Butterworth-Heinemann; 2014.
- [51] Pilkey WD. Analysis and design of elastic beams computational methods; 1979. <http://doi.org/10.1177/058310247901101112>.
- [52] Timoshenko SP, Woinowsky-Krieger S. Theory of plates and shells. McGraw-hill; 1959.
- [53] Vlasov VZ. Thin-Walled-Elastic-Beams; 1961.
- [54] Ota NSN, Wilson L, Gay Neto A, Pellegrino S, Pimenta PM. Nonlinear dynamic analysis of creased shells. *Finite Elem Anal Des* 2016;121:64–74. <https://doi.org/10.1016/j.finel.2016.07.008>.
- [55] Love AEH. The small free vibrations and deformation of a thin elastic shell. *Philos Trans R Soc Lond A* 1888;179:491–546.
- [56] Kirchhoff G. Über das Gleichgewicht und die Bewegung einer elastischen Scheibe; 1826. p. 51–88.
- [57] Reissner E. Stress strain relations in the theory of elastic shells. *J Math Phys* 1952;31:109–19.
- [58] Donnel LH. Stability of thin-walled tubes under torsion. USA: Pasadena; 1933.
- [59] Sanders JL. An improved first-approximation theory for thin shells; 1959. p. 1–11. doi:TR R-24.
- [60] Flügge W. General properties of stress systems in shells. *Stress. Shells* Berlin, Heidelberg: Springer; 1960. p. 1–18. https://doi.org/10.1007/978-3-662-01028-0_1.
- [61] Mushtari KM, Galimov KZ. Non-linear theory of thin elastic shells. *Academy of Sciences: U.S.S.R. Kazan' Branch*; 1957.
- [62] Sanders Jr JL. Nonlinear theories for thin shells; 1961.
- [63] Naghdi PM, Nordgren RP. On the nonlinear theory of elastic shells under the Kirchhoff hypothesis. *Q Appl Math* 1963;21:49–59.
- [64] Simo JC, Rifai MS, Fox DD. On a stress resultant geometrically exact shell model. Part VI: Conserving algorithms for non-linear dynamics. *Int J Numer Methods Eng* 1992;34:117–64. <https://doi.org/10.1002/nme.1620340108>.
- [65] Campello EMB, Pimenta PM, Wriggers P. A triangular finite shell element based on a fully nonlinear shell formulation. *Comput Mech* 2003;31:505–18. <https://doi.org/10.1007/s00466-003-0458-8>.
- [66] Shama M. Torsion and shear stresses in ships. Berlin, Heidelberg: Springer; 2011. <http://doi.org/10.1007/978-3-642-14633-6>.
- [67] Prokić A. Computer program for determination of geometrical properties of thin-walled beams with open-closed section. *Comput Struct* 2000;74:705–15. [https://doi.org/10.1016/S0045-7949\(99\)00076-0](https://doi.org/10.1016/S0045-7949(99)00076-0).
- [68] Lee J. Center of gravity and shear center of thin-walled open-section composite beams. *Compos Struct* 2001;52:255–60. [https://doi.org/10.1016/S0263-8223\(00\)00177-X](https://doi.org/10.1016/S0263-8223(00)00177-X).
- [69] Gay Neto A. Generic interface readily accessible for finite elements; 2016, p. 1–70.
- [70] Saito MT, Gay Neto A. Computational simulation of woven fabrics using beam-to-beam contact formulation. In: CILAMCE 2016, Brasília; 2016.
- [71] Pereira TFM, Gay Neto A. Computational model to evaluate actions in railwheel contact interaction. In: CILAMCE 2016, Brasília; 2016.
- [72] Faccio Junior CJ, Gay Neto A. Dynamics of wind turbine blades using a geometrically-exact beam formulation. In: ECCOMAS Congr. 2016, Crete Island, Greece; 2016. p. 5–10.
- [73] Jonkman J. NREL's S809 Airfoil Graphic and Coordinates; (n.d.).
- [74] Somers DM. Design and experimental results for the S809 airfoil, vol. 104; 1997. <http://doi.org/10.2172/437668>.
- [75] Bir G. User's Guide to PreComp (Pre-Processor for Computing Composite Blade Properties); 2006, p. 31.
- [76] Jones RM. Mechanics of composite materials. *Mech Compos Mater* 1999;519. <https://doi.org/10.1007/BF00611782>.
- [77] de Campos PR, Gay Neto A. Rigid body formulation in a finite element context with contact interaction. *Comput Mech* 2018. <https://doi.org/10.1007/s00466-018-1569-6>.